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Komori et al.

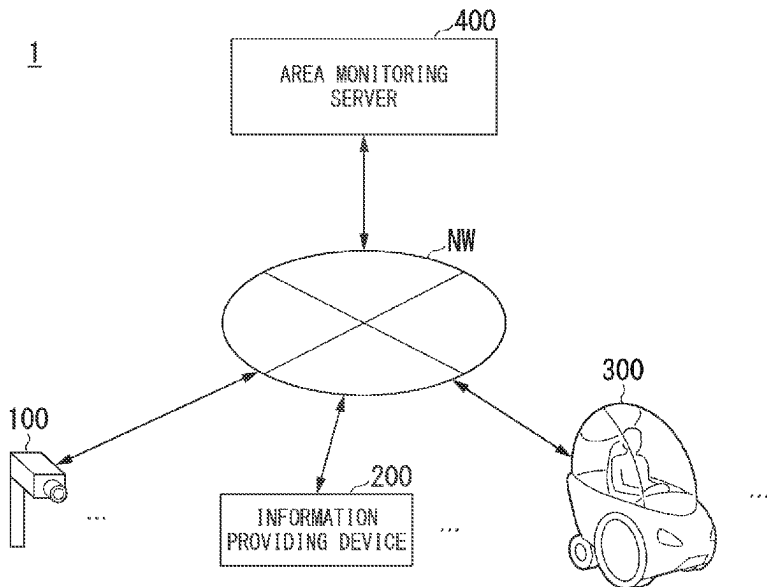
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(54) **AREA MONITORING SYSTEM AND AREA MONITORING METHOD**
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G08G 1/04 (2006.01)
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(52) **U.S. Cl.**
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(58) **Field of Classification Search**
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(57) **ABSTRACT**
Area monitoring system includes a plurality of sensor devices each installed in a monitoring target area and configured to detect a mobile object moving in the monitoring target area and a determiner configured to determine whether or not mobile objects detected by the plurality of sensor devices are the same physical object on the basis of detection results of the plurality of sensor devices. The plurality of sensor devices include at least a radio wave sensor and an optical sensor. The determiner determines whether or not mobile objects detected by the radio wave sensor and the optical sensor are the same physical object on the basis of information of at least one of feature quantities regarding movements of the mobile objects and reflection intensities corresponding to attributes of the mobile objects when the mobile objects have been detected by the radio wave sensor and the optical sensor.

10 Claims, 14 Drawing Sheets



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G08G 1/16 (2006.01)

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CPC B60Q 5/00; B60Q 9/00; B60R 1/24; B60R
1/29; B60R 1/30; B60R 1/31; B60R
16/02; G06V 20/52; G06V 20/56

USPC 701/117

See application file for complete search history.

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FIG. 1

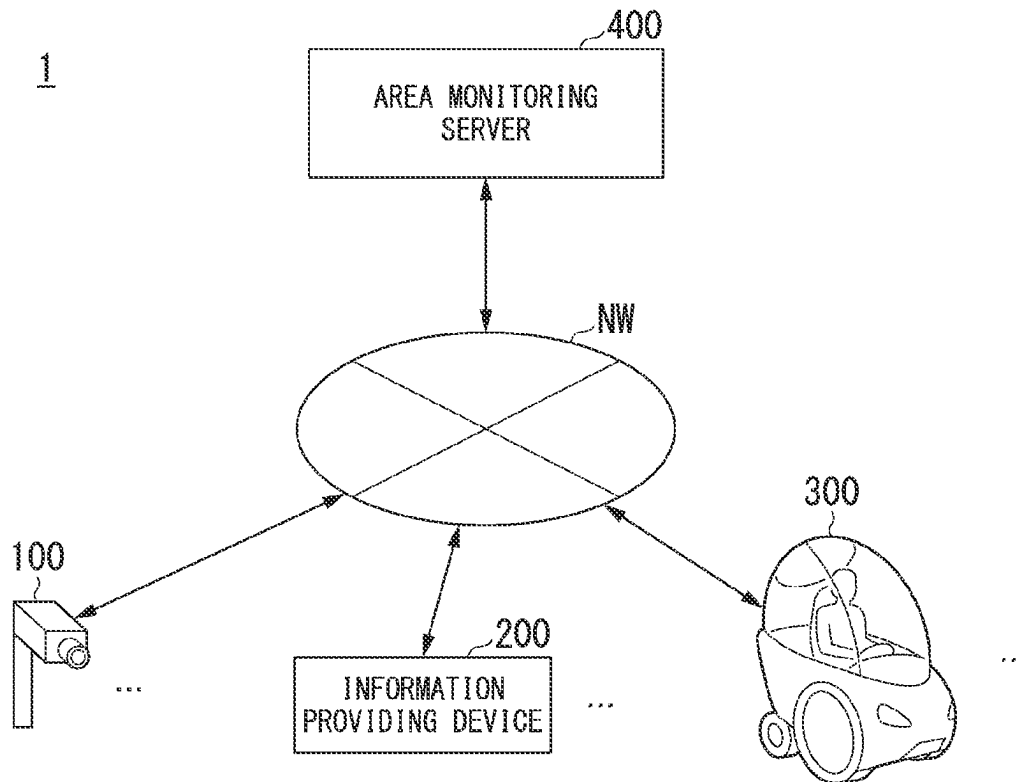


FIG. 2

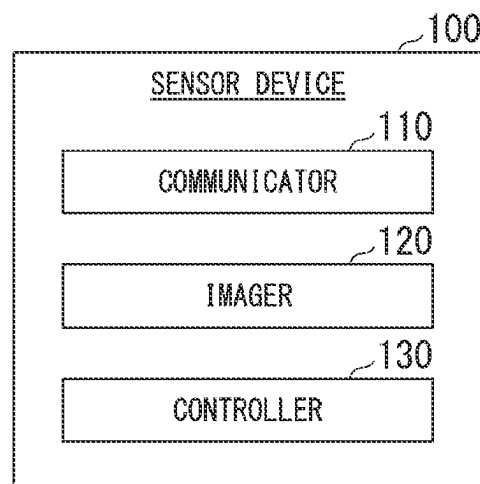


FIG. 3

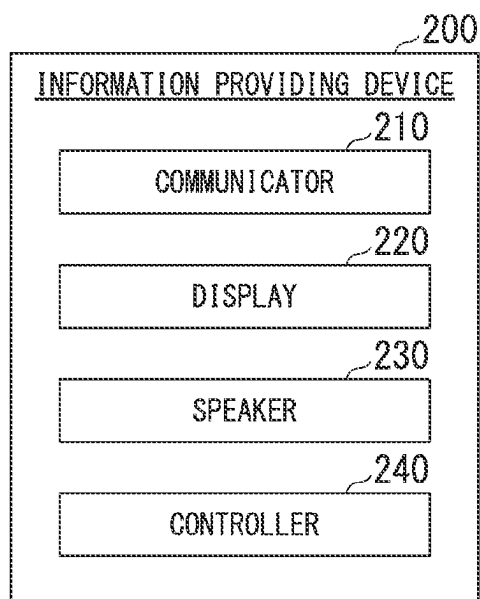


FIG. 4

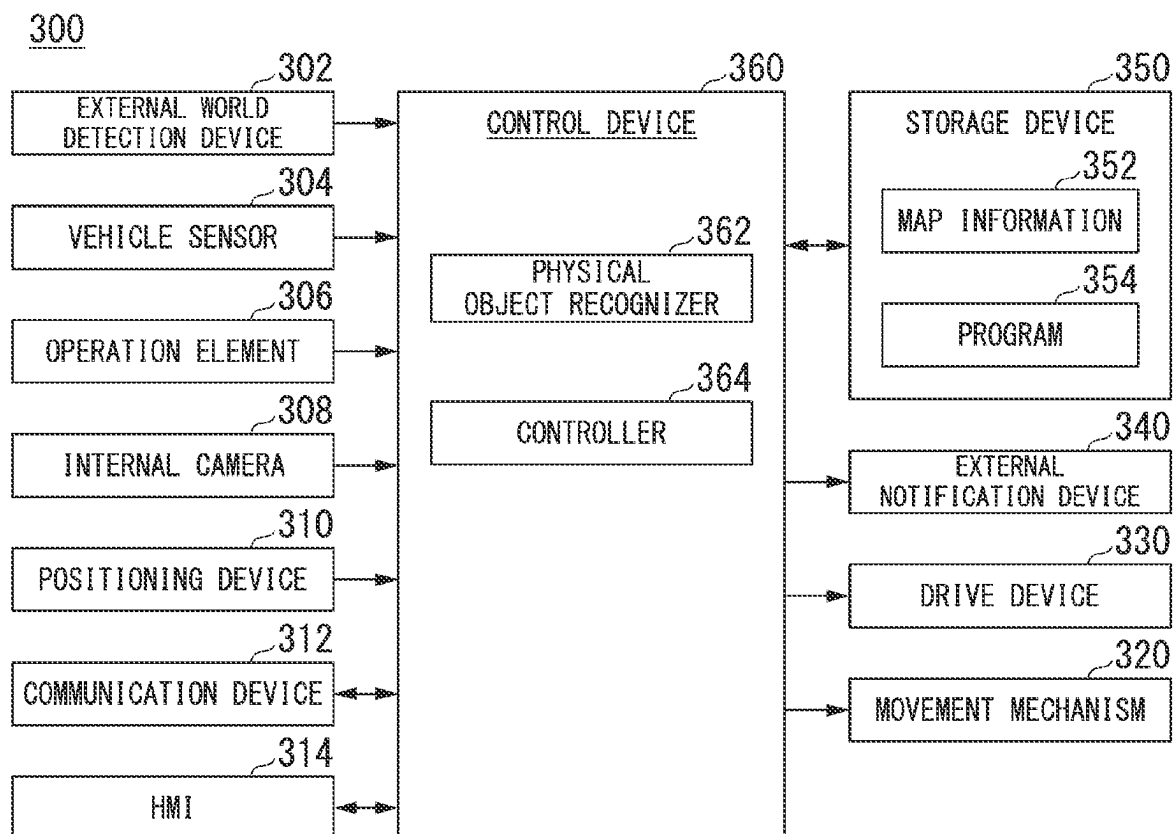


FIG. 5

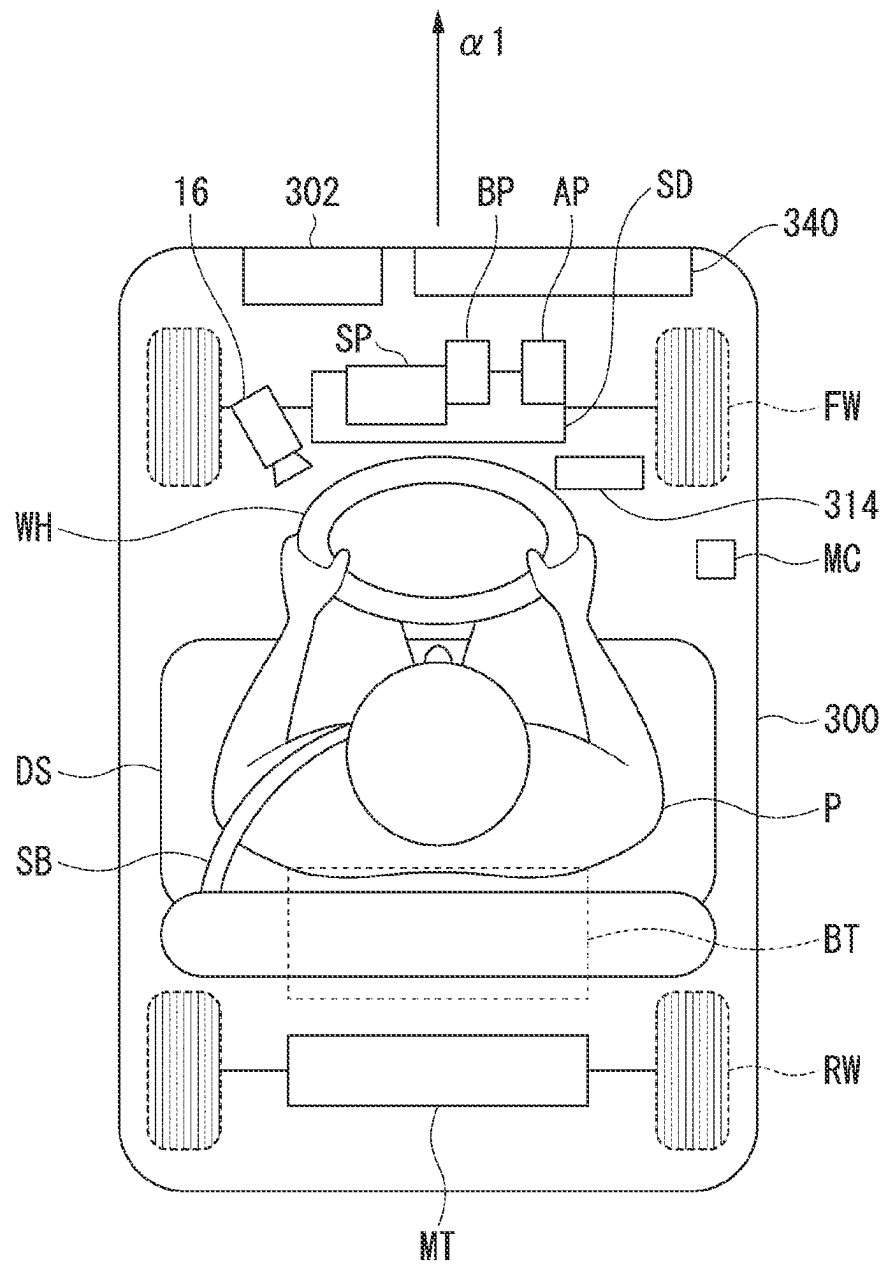


FIG. 6

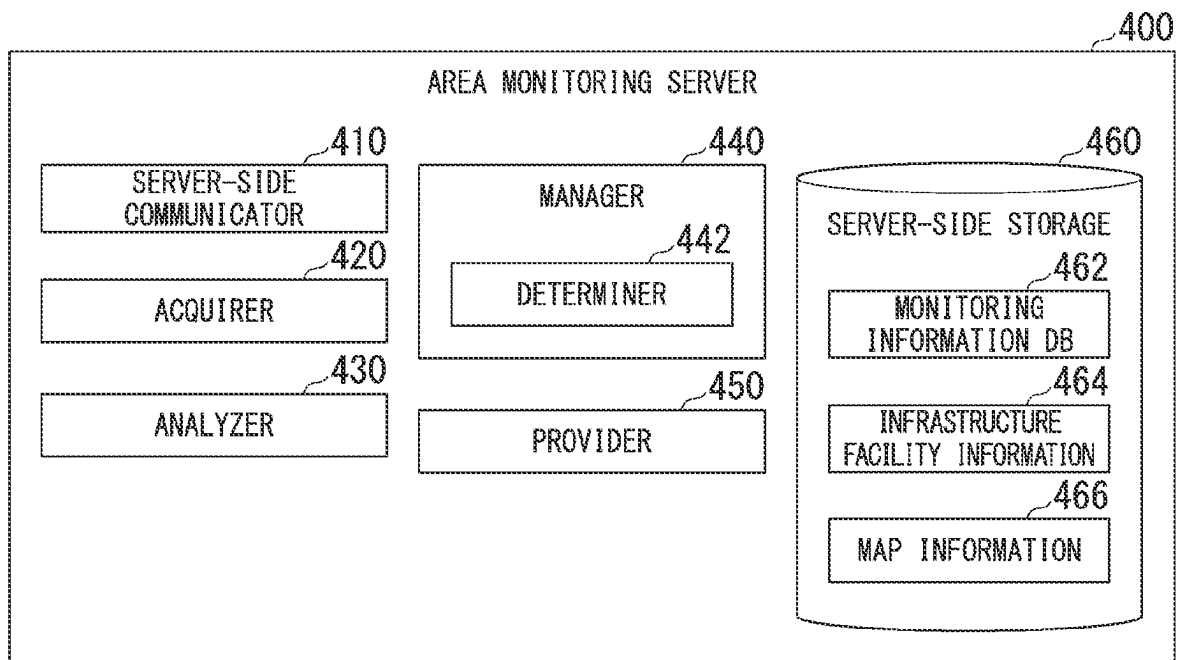


FIG. 7

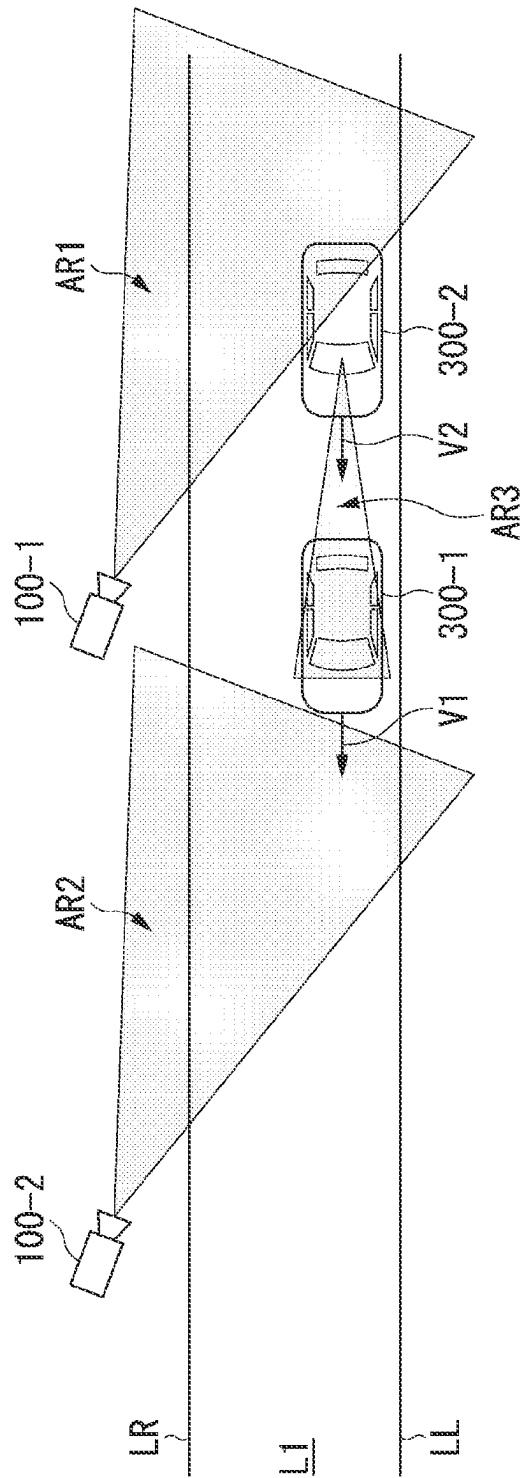


FIG. 8

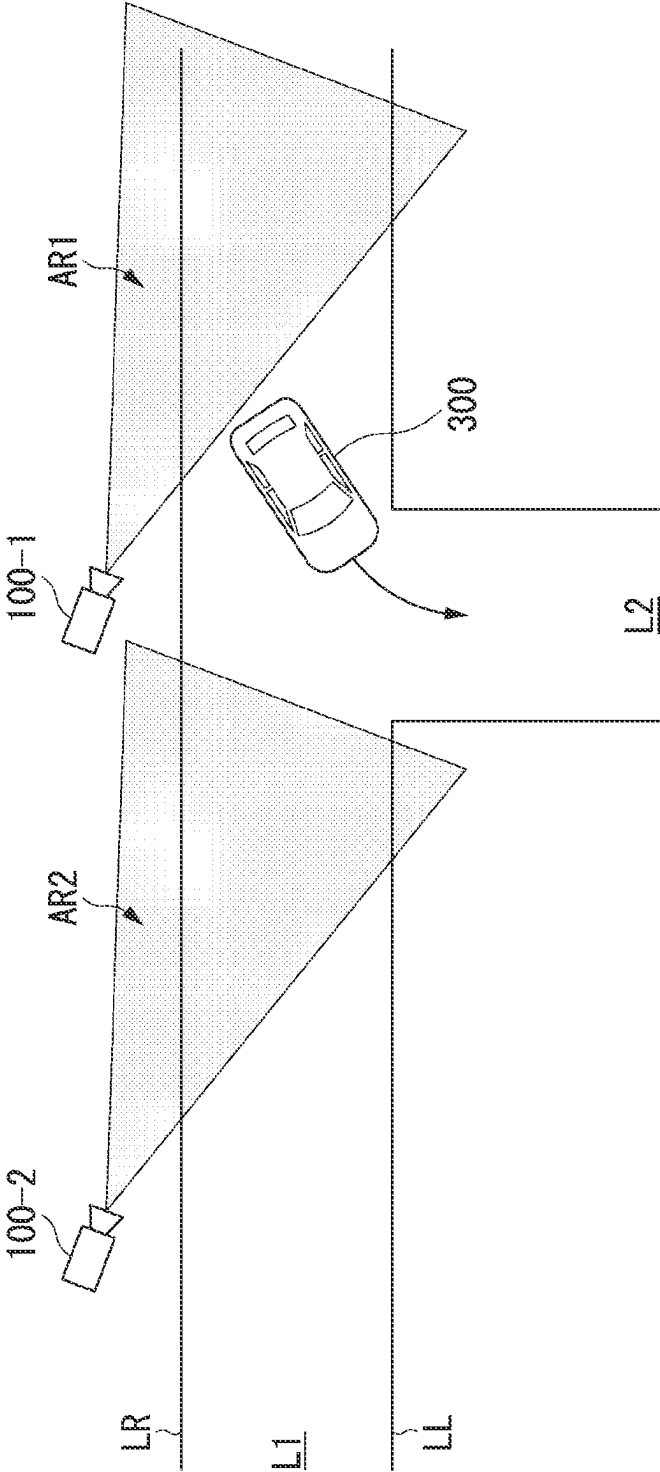


FIG. 9

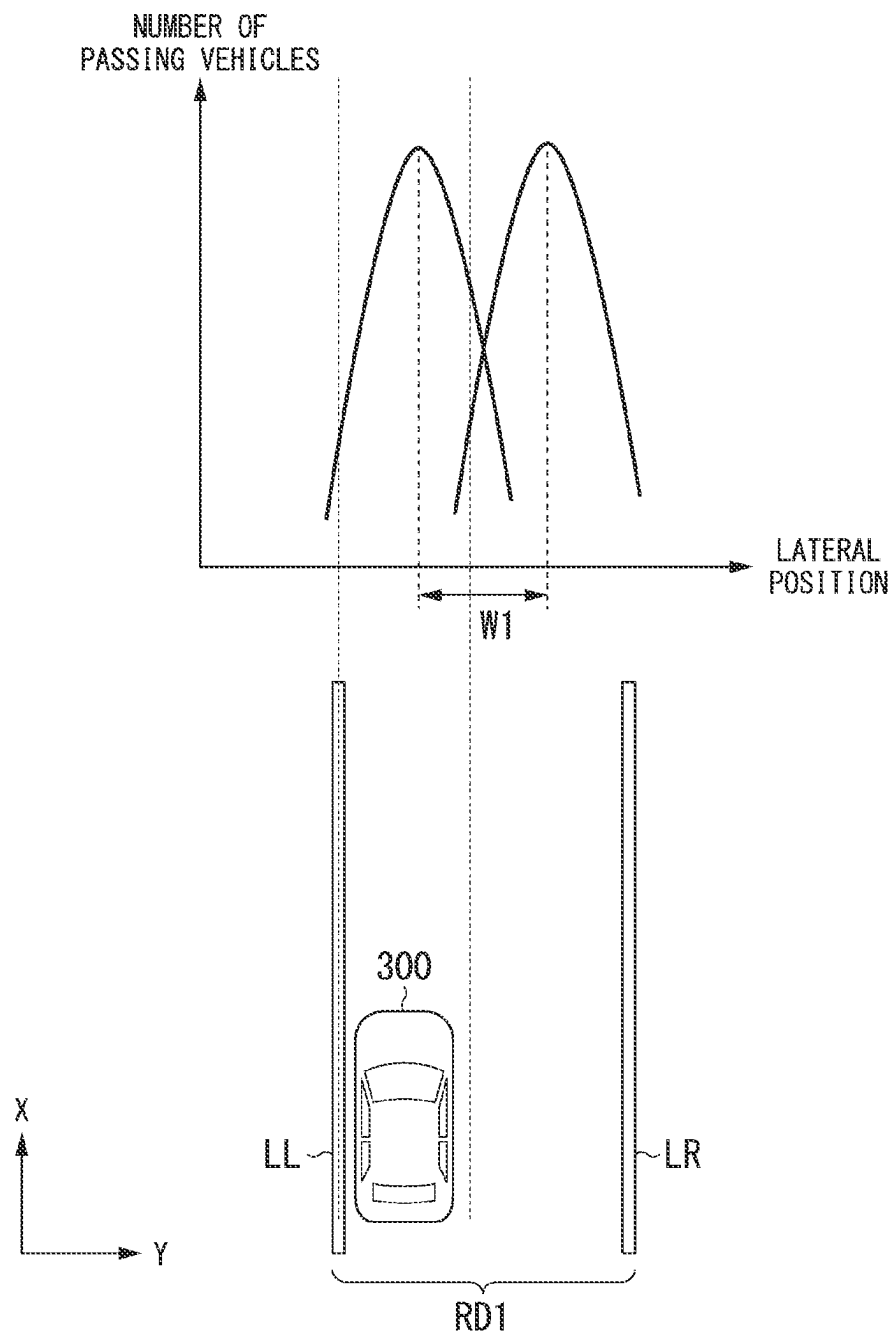


FIG. 10

SENSOR TYPE	POSITION ACCURACY	SPEED ACCURACY	DETECTION RANGE	SITUATION ASCERTAINMENT	ENVIRONMENTAL RESISTANCE
STEREO CAMERA	⊙	○	○	⊙	△
MONOCULAR CAMERA	○	○	○	⊙	△
FISHEYE CAMERA	△	△	△	⊙	△
INFRARED CAMERA	○	○	○	⊙	○
TOF CAMERA	⊙	○	⊙	△	△
RADAR DEVICE	○	⊙	⊙	△	⊙
LIDAR SENSOR	⊙	○	△	○	○
MICROPHONE	△	△	△	△	○

FIG. 11

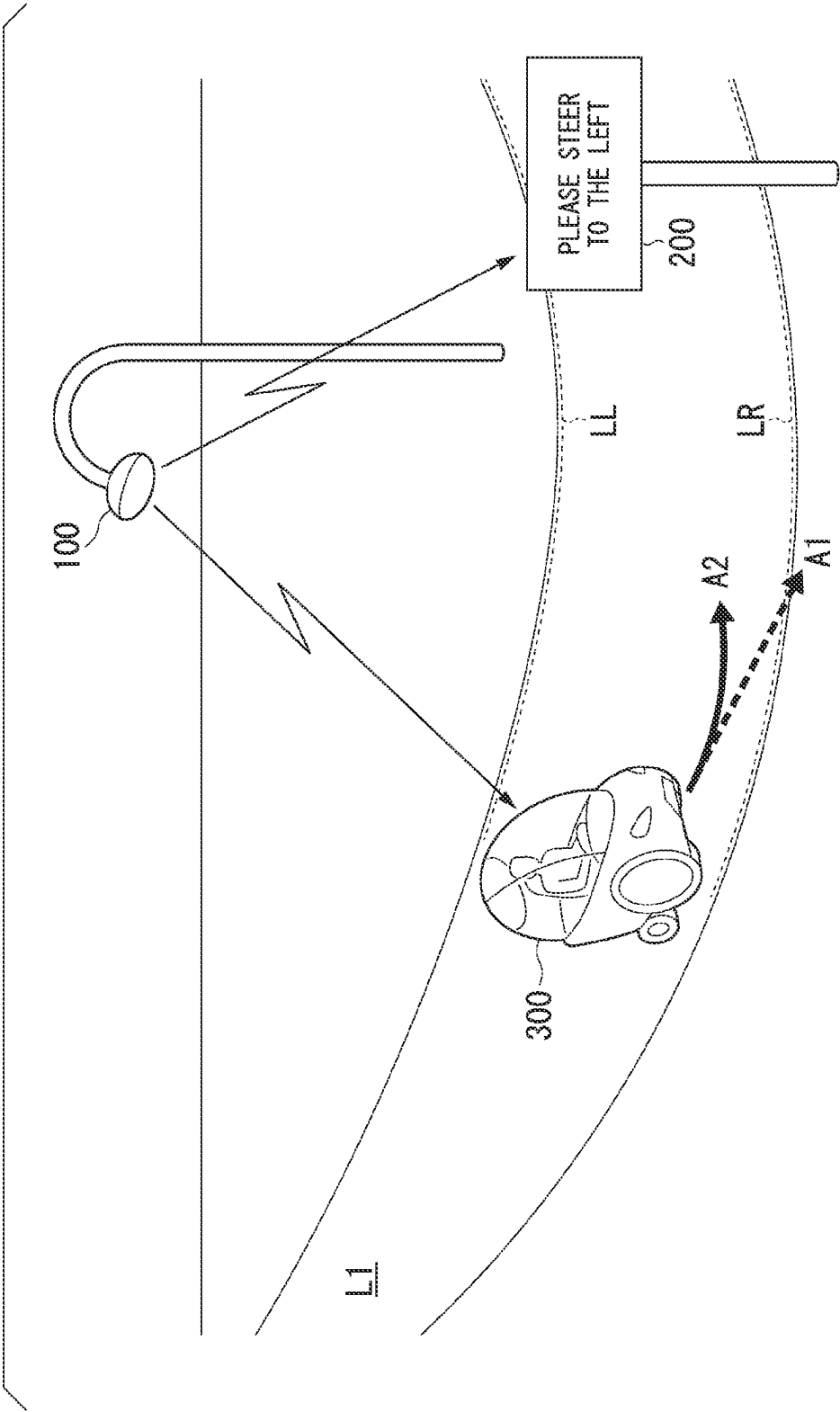
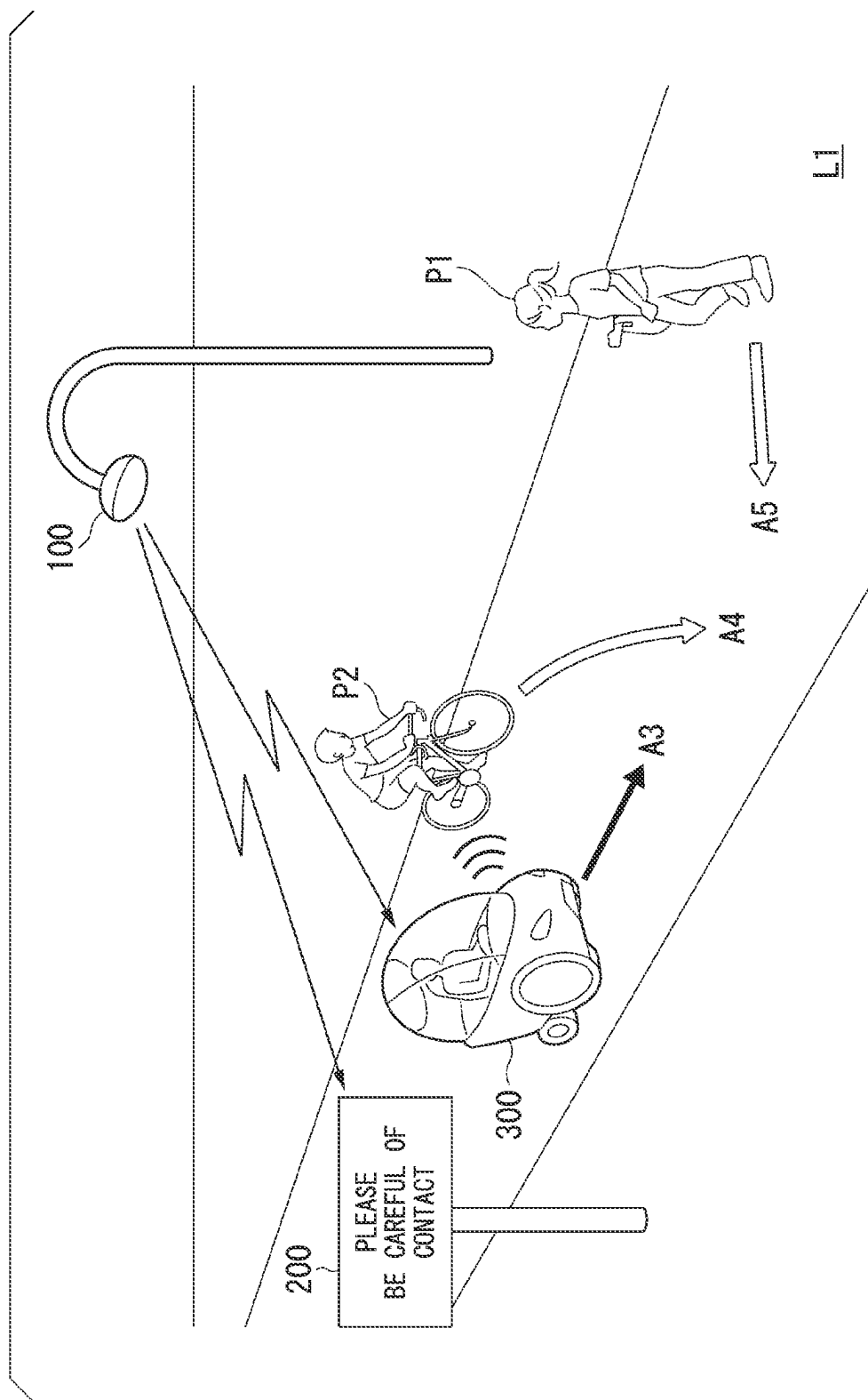


FIG. 12



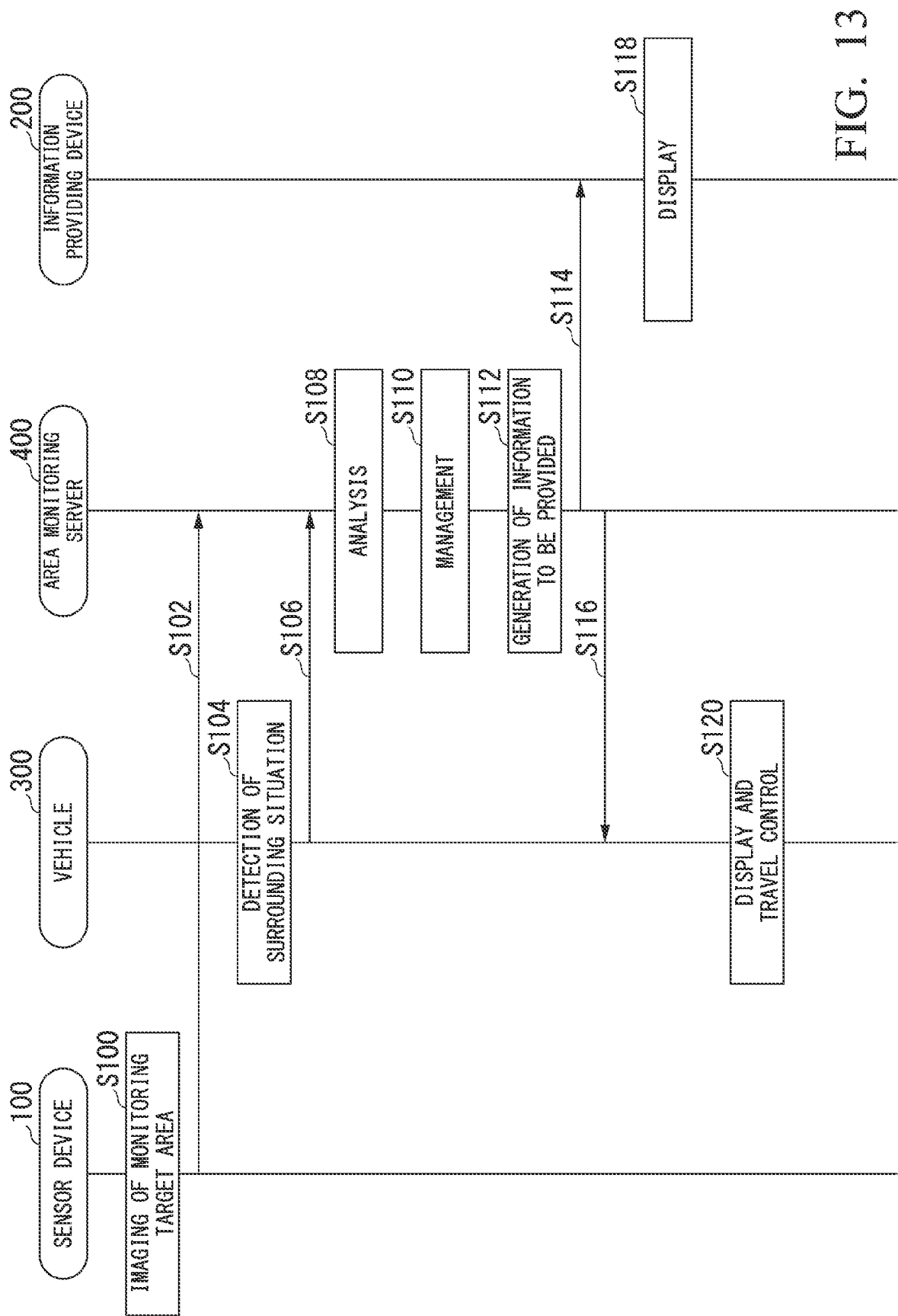


FIG. 13

FIG. 14

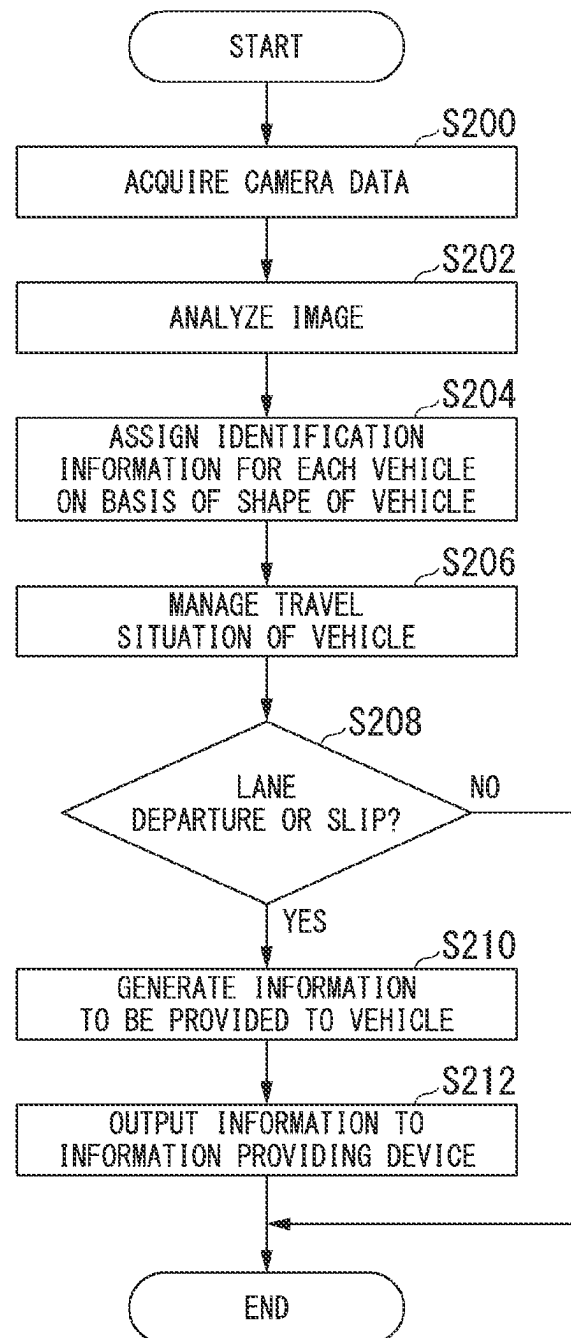


FIG. 15

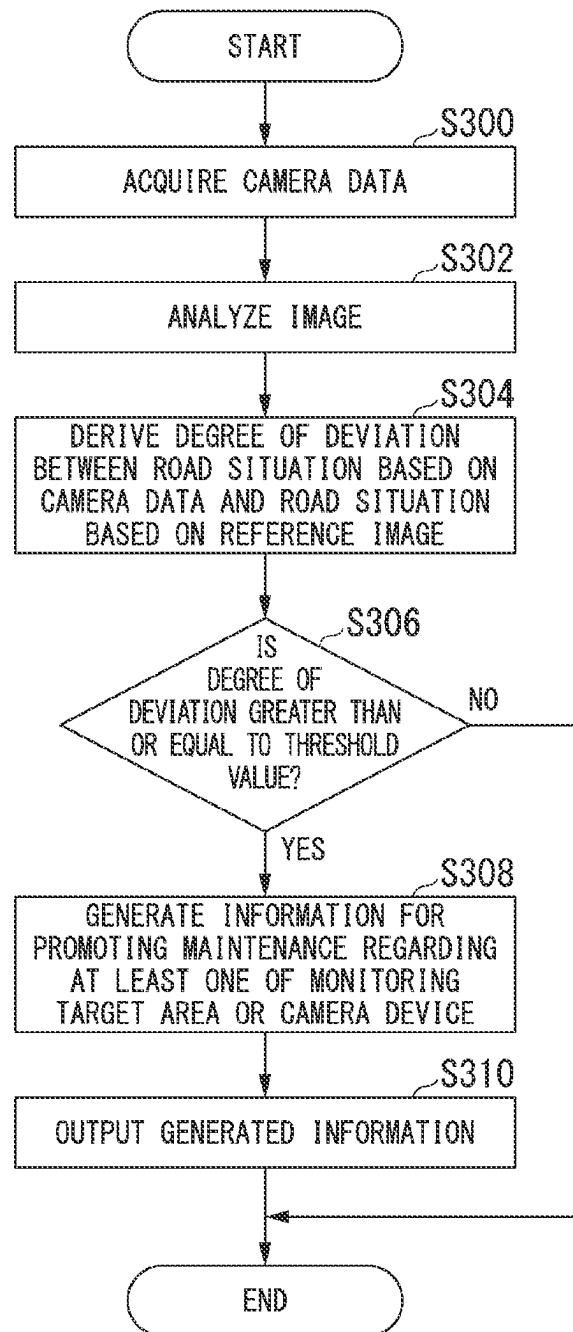
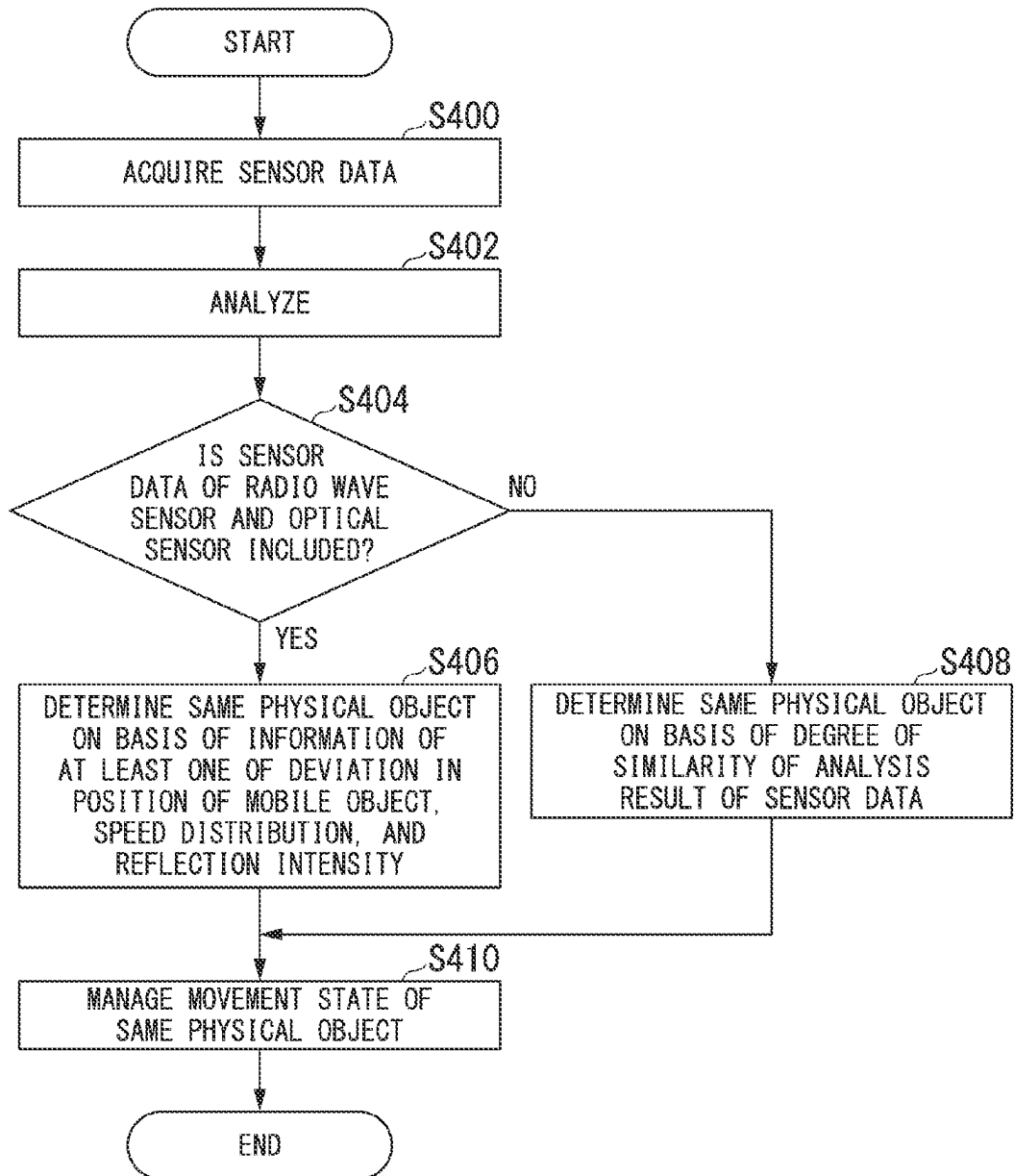


FIG. 16



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AREA MONITORING SYSTEM AND AREA MONITORING METHOD

CROSS-REFERENCE TO RELATED APPLICATION

Priority is claimed on Japanese Patent Application No. 2022-158274, filed Sep. 30, 2022, the content of which is incorporated herein by reference.

BACKGROUND

Field of the Invention

The present invention relates to an area monitoring system and an area monitoring method.

Description of Related Art

In recent years, efforts to provide access to sustainable transportation systems that take into account vulnerable groups among traffic participants have become active. To implement these, research and development for further improving traffic safety and convenience through research and development relating to safety precaution technology are being focused on. In relation to this, conventionally, technology for outputting a warning when a vehicle departs from a travel lane using a sensor mounted in the vehicle or determining the start of movement of a physical object using detection results of a plurality of detectors is known (for example, Japanese Unexamined Patent Application, First Publication Nos. 2014-31167 and 2021-51466).

SUMMARY

Meanwhile, in the technology relating to safety precautions, a state of a mobile object moving within an area or a state of a road may not be accurately ascertained due to changes in a situation relating to a monitoring target area and the like. Therefore, there is a possibility that it will be impossible to appropriately monitor the monitoring target area.

In order to solve the above problems, an objective of the present application is to provide an area monitoring system and an area monitoring method capable of more appropriately monitoring a monitoring target area. In turn, it will contribute to the development of a sustainable transportation system.

An area monitoring system and an area monitoring method according to the present invention adopt the following configurations.

(1): According to an aspect of the present invention, there is provided an area monitoring system including: a plurality of sensor devices each installed in a monitoring target area and configured to detect a mobile object moving in the monitoring target area; and a determiner configured to determine whether or not mobile objects detected by the plurality of sensor devices are the same physical object, wherein the plurality of sensor devices include at least a radio wave sensor and an optical sensor, and wherein the determiner determines whether or not mobile objects detected by the radio wave sensor and the optical sensor are the same physical object on the basis of information of at least one of feature quantities regarding movements of the mobile objects and reflection intensities corresponding to

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attributes of the mobile objects when the mobile objects have been detected by the radio wave sensor and the optical sensor.

(2): In the above-described aspect (1), the determiner determines whether or not the mobile objects are the same physical object on the basis of information of at least one of deviations in positions of the mobile objects, speed distributions, and reflection intensities corresponding to the attributes of the mobile objects when the mobile objects move outside of a sensor range after being detected by the optical sensor and the radio wave sensor is located at a movement destination.

(3): In the above-described aspect (1), the determiner determines whether or not the mobile objects are the same physical object on the basis of information of at least one of deviations in positions of the mobile objects, speed distributions, and reflection intensities corresponding to the attributes of the mobile objects when the mobile objects move outside of a sensor range after being detected by the radio wave sensor and the optical sensor is located at a movement destination.

(4): In the above-described aspect (1), the area monitoring system further includes a manager configured to manage a situation of the monitoring target area, wherein, when the mobile object is a vehicle and there is a physical object approaching the vehicle, the manager issues an action instruction for the vehicle.

(5): In the above-described aspect (4), the manager differentiates content of the action instruction for the vehicle in accordance with a degree of influence on contact of a physical object approaching the vehicle.

(6): In the above-described aspect (4), when an information providing device is set in the monitoring target area, the manager causes an image or sound indicating content of the action instruction to be output to the information providing device.

(7): In the above-described aspect (6), when an image sensor is included in the plurality of sensor devices, the manager causes information about an attribute of a physical object approaching the vehicle included in an image captured by the image sensor to be output to the information providing device.

(8): In the above-described aspect (4), the manager outputs information for causing the vehicle to execute one or both of speed control and steering control corresponding to content of the action instruction for the vehicle to the vehicle.

(9): In the above-described aspect (1), the radio wave sensor is installed in a segment where curvature of a road included in the monitoring target area is less than a threshold value and the optical sensor is installed in a segment where the curvature of the road is greater than or equal to the threshold value.

(10): According to another aspect of the present invention, there is provided an area monitoring method including: determining, by a computer, whether or not mobile objects detected by a plurality of sensor devices each installed in a monitoring target area and configured to detect a mobile object moving in the monitoring target area are the same physical object on the basis of detection results of the plurality of sensor devices, wherein the plurality of sensor devices include at least a radio wave sensor and an optical sensor, and determining, by the computer, whether or not mobile objects detected by the radio wave sensor and the optical sensor are the same physical object on the basis of information of at least one of feature quantities regarding movements of the mobile objects and reflection intensities

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corresponding to attributes of the mobile objects when the mobile objects have been detected by the radio wave sensor and the optical sensor.

According to the above-described aspects (1) to (10), it is possible to more appropriately monitor a monitoring target area.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram showing an example of a configuration of an area monitoring system according to an embodiment.

FIG. 2 is a diagram showing an example of a configuration of a sensor device.

FIG. 3 is a diagram showing an example of a configuration of an information providing device.

FIG. 4 is a diagram showing an example of a configuration of a vehicle.

FIG. 5 is a perspective view of a vehicle seen from above.

FIG. 6 is a diagram showing an example of a configuration of an area monitoring server.

FIG. 7 is a diagram for describing a first monitoring example.

FIG. 8 is a diagram for describing a second monitoring example.

FIG. 9 is a diagram for describing a third monitoring example.

FIG. 10 is a diagram showing an example of characteristic information for a sensor type.

FIG. 11 is a diagram showing a first example of information provision.

FIG. 12 is a diagram showing a second example of information provision.

FIG. 13 is a sequence chart showing an example of a flow of a process executed by the area monitoring system of the embodiment.

FIG. 14 is a flowchart showing an example of a first process.

FIG. 15 is a flowchart showing an example of a second process.

FIG. 16 is a flowchart showing an example of a third process.

DESCRIPTION OF EMBODIMENTS

Hereinafter, embodiments of an area monitoring system and an area monitoring method of the present invention will be described with reference to the drawings. Mobile objects in the area monitoring system of the embodiment include a physical object capable of moving on a road such as a vehicle with two, three, or four wheels or the like, a bicycle, or a person (a pedestrian). "Roads" may include not only roads (lanes) dedicated to vehicles, but also roads, passages, road surfaces, areas having a prescribed size and length in which vehicles and people can move, and the like. Vehicles includes any vehicle capable of being boarded by a person (a driver) and moving on the road surface, for example, a single-seater vehicle (micromobility) and the like. In the following description, the vehicle will be described as four-wheeled micromobility. Although a case in which left-hand traffic regulations are applied will be described below, it is only necessary to reverse the left and right when right-hand traffic regulations are applied.

[System Configuration]

FIG. 1 is a diagram showing an example of a configuration of an area monitoring system 1 according to an embodiment. The area monitoring system 1 shown in FIG. 1

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includes, for example, a sensor device 100, an information providing device 200, a vehicle 300, and an area monitoring server 400. These are communicatively connected, for example, via a network NW. The network NW includes, for example, the Internet, a cellular network, a Wi-Fi network, a wide area network (WAN), a local area network (LAN), a provider device, a wireless base station, and the like. The area monitoring system 1 may include one or more of the sensor device 100, the information providing device 200, the vehicle 300, and the area monitoring server 400. The area monitoring system 1 may have a configuration that does not include at least one of the information providing device 200 or the vehicle 300. The area monitoring system 1 may transmit and receive information to and from each configuration via one or more relay devices (for example, gateway devices and small servers).

The sensor device 100 detects a physical object located in the monitoring target area. The monitoring target area is, for example, an area including a road where a mobile object such as a traffic participant such as the vehicle 300, a pedestrian, or a bicycle passes. A plurality of sensor devices 100 are installed at prescribed intervals along a road or the like when the monitoring target area is a road or the like. The sensor device 100 may include an optical sensor and a radio wave sensor. The optical sensor is, for example, a camera device (an image sensor) such as a digital camera. Specifically, optical sensors include a stereo camera, a monocular camera, a fisheye camera, an infrared camera, and the like. Radio wave sensors include a radar device, a light detection and ranging (LIDAR) sensor, a time of flight (TOF) camera, and the like. The radar device emits radio waves such as millimeter waves around the sensor device 100 and detects radio waves (reflected waves) reflected by the physical object to detect at least a position (distance and direction) of the physical object. The LIDAR sensor applies light (or electromagnetic waves having a wavelength close to that of light) around the sensor device 100 and measures scattered light. The LIDAR sensor detects a distance from the sensor device 100 to the target on the basis of a period of time from light emission to light reception. The applied light is, for example, pulsed laser light. When the monitoring target area is a road, the sensor device 100, for example, is installed to image an area including a road from above the road. Sensor ranges (imaging ranges) of the plurality of sensor devices 100, which are installed, may or may not overlap at least partly.

The information providing device 200 is installed near the monitoring target area and provides information to the vehicle 300 and traffic participants passing through the monitoring target area. The information providing device 200 may be a terminal device (for example, a smartphone or a tablet terminal) owned by a manager (a maintenance staff member) who maintains (manages) the monitoring target area.

The vehicle 300, for example, travels with motive power output in accordance with an operation of an internal combustion engine operating with fuel such as gasoline, diesel oil, or hydrogen. The vehicle 300 may travel with an electric motor driven by electric power supplied from a battery. The battery is, for example, a lithium-ion secondary battery (LIB), a nickel-metal hydride battery, an all-solid-state battery, or the like. The vehicle 300 may be a hybrid vehicle using the above-described internal combustion engine or electric motor as a drive source. An internal combustion engine or electric motor is an example of a power source mounted in the vehicle 300.

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The area monitoring server **400** monitors a situation of a monitoring target area (for example, a situation of a mobile object moving within the area, a situation of a road surface, or an installation situation of the sensor device **100**) on the basis of information obtained from the sensor device **100** or the vehicle **300** via the network NW and causes information based on a monitoring result to be output through the information providing device **200**. The area monitoring server **400** may be implemented, for example, in a server device or a storage device incorporated in a cloud computing system. In this case, the function of the area monitoring server **400** may be implemented by a plurality of server devices or a plurality of storage devices in the cloud computing system.

Next, functional configurations of the sensor device **100**, the information providing device **200**, the vehicle **300**, and the area monitoring server **400** will be specifically described.

[Sensor Device]

FIG. **2** is a diagram showing an example of a configuration of the sensor device **100**. In the example of FIG. **2**, a configuration when the sensor device **100** is a camera device is shown. The sensor device **100** includes, for example, a communicator **110**, an imager **120**, and a controller **130**.

The communicator **110** communicates with the area monitoring server **400** and other external devices via the network NW. For example, the communicator **110** transmits captured image data from the imager **120** to the area monitoring server **400**. The communicator **110** may transmit information received from the area monitoring server **400** to the information providing device **200** or the vehicle **300** located in a nearby area.

The imager **120** is a digital camera using, for example, a solid-state image sensor such as a charge-coupled device (CCD) or a complementary metal oxide semiconductor (CMOS). The imager **120** iteratively images an area including the monitoring target area at a prescribed interval and at a prescribed timing. An angle of view (an imaging area) of the imager **120** is fixed. The imager **120** may be a stereo camera, a monocular camera, a fisheye camera, an infrared camera, or the like.

When the sensor device **100** is a radio wave sensor, a radar device, a LIDAR sensor, a TOF camera, or the like is provided instead of (or in addition to) the imager **120**. A microphone configured to collect an ambient sound may be provided in the sensor device **100**.

The controller **130** controls an overall configuration of the sensor device **100**. For example, the controller **130** transmits sensor data including an image captured by the imager **120** (hereinafter referred to as a camera image), imaging date and time information, and identification information (for example, a sensor ID) for identifying the sensor device **100** to the area monitoring server **400** via the network NW. The sensor data may include information obtained by various types of sensors such as a radar device, a LIDAR sensor, and a microphone in addition to (or instead of) a camera image.

FIG. **3** is a diagram showing an example of a configuration of the information providing device **200**. The information providing device **200** includes, for example, a communicator **210**, a display **220**, a speaker **230**, and a controller **240**.

The communicator **210** communicates with the area monitoring server **400** and other external devices via the network NW. For example, the communicator **210** receives various types of information transmitted from the area monitoring server **400**.

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The display **220** displays an image relating to information provided by the area monitoring server **400**. The display **220** is, for example, a digital signage system such as an electronic bulletin board or an electronic signboard. The speaker **230** outputs a sound relating to the information provided from the area monitoring server **400**. It is only necessary for the information providing device **200** to include at least one of the display **220** and the speaker **230**.

The controller **240** controls an overall configuration of the information providing device **200**. For example, the controller **240** generates an image or a sound relating to the information provided by the area monitoring server **400** and causes the generated image or sound to be output from the display **220** or the speaker **230**. When an image or sound is provided from the area monitoring server **400**, the controller **240** may cause the image or sound to be output to the display **220** or the speaker **230** as it is.

A light emitter may be provided in the information providing device **200** instead of (or in addition to) the configuration described above. The light emitter turns on or flashes, for example, a light emitting body provided at least in a part of a road surface marking such as a stop line or a pedestrian crossing provided on a road in the monitoring target area. The light emitter is, for example, a light emitting diode (LED), but is not limited thereto. The light emitter may emit light around the display or at least a part thereof. The light emitter may emit light in a prescribed color or may emit light in a color indicated by the controller **240**. The controller **240** causes the light emitter to emit light according to an instruction from the area monitoring server **400**. The light emitter is provided, for example, at a pedestrian crossing for crossing a road or a stop line for stopping the movement of a vehicle.

[Vehicle]

FIG. **4** is a diagram showing an example of a configuration of the vehicle **300**. The vehicle **300** includes, for example, an external world detection device **302**, a vehicle sensor **304**, an operation element **306**, an internal camera **308**, a positioning device **310**, a communication device **312**, a human-machine interface (HMI) **314**, a movement mechanism **320**, a drive device **330**, an external notification device **340**, a storage device **350**, and a control device **360**. Among these components, some components that are not essential for implementing the functions of the present invention may be omitted.

The external world detection device **302** detects an external situation of the vehicle **300**. For example, the external world detection device **302** includes various types of devices having a detection range of at least a part (including a travel direction) around the vehicle **300**. The external world detection device **302** includes an external camera, a radar device, a LIDAR sensor, a sensor fusion device, and the like. The external camera is, for example, a digital camera using a solid-state image sensor such as a charge-coupled device (CCD) or a complementary metal oxide semiconductor (CMOS). The external camera is attached to any location in the vehicle **300**. When the view in front of the vehicle is imaged, the external camera is attached to an upper part of a front windshield, a rear surface of a rearview mirror, or the like. The external camera periodically and iteratively images the surroundings (surroundings including the travel direction) of the vehicle **300**. The external camera may be a stereo camera, a monocular camera, a fisheye camera, or the like.

The radar device radiates radio waves such as millimeter waves around the vehicle **300** and detects at least a position (a distance to and a direction) of a physical object by detecting radio waves (reflected waves) reflected by the

physical object. The radar device is attached to any location on the vehicle **300**. The vehicle **300** may detect a position and a speed of the physical object in a frequency-modulated continuous wave (FM-CW) scheme. The LIDAR sensor radiates light (or electromagnetic waves having a wavelength close to light) to the vicinity of the vehicle **300** and measures scattered light. The LIDAR sensor detects a distance from the vehicle **300** to a target on the basis of a period of time from light emission to light reception. The LIDAR sensor is attached to any location on the vehicle **300**. The external world detection device **302** outputs information indicating a detection result (an image, a position of a physical object, or the like) to the control device **360**.

The vehicle sensor **304** includes, for example, a speed sensor, an acceleration sensor, a yaw rate (angular rate) sensor, a direction sensor, an operation amount detection sensor attached to the operation element **306**, and the like.

The operation element **306** accepts a driving operation from an occupant of the vehicle **300**. The operation element **306** includes, for example, an operation element (for example, an accelerator pedal, a brake pedal, a dial switch, or a lever) for issuing an acceleration/deceleration instruction and an operation element (for example, a steering wheel) for issuing a steering instruction. In this case, the vehicle sensor **304** may include an accelerator opening degree sensor, a brake depression amount sensor, a steering torque sensor, and the like. The vehicle **300** may include an operation element of a form other than the above (for example, a non-annular rotation operation element, a joystick, a button, or the like) as the operation element **306**.

The internal camera **308** images at least the head of the occupant of the vehicle **300** from the front. The internal camera **308** is a digital camera that uses an imaging element such as a CCD or a CMOS. The internal camera **308** outputs the captured image to the control device **360**.

The positioning device **310** is a device for measuring a position of the vehicle **300**. The positioning device **310** is, for example, a global navigation satellite system (GNSS) receiver and identifies the position of the vehicle **300** on the basis of a signal received from a GNSS satellite and outputs the identified position as position information. Also, the position information of the vehicle **300** may be estimated from a position of a Wi-Fi base station to which the communication device **312** is connected. The positioning device **310** may include the vehicle sensor **304**.

The communication device **312** communicates with another vehicle located in a nearby area using, for example, a cellular network, a Wi-Fi network. Bluetooth (registered trademark), dedicated short-range communication (DSRC), or the like, or communicates with various types of external devices (for example, the area monitoring server **400**, the sensor device **100**, and the information providing device **200**) via a radio base station.

The HMI **314** presents various types of information (or provides a notification) to the occupant of the vehicle **300** and receives an input operation from the occupant. The HMI **314** includes various types of display devices, a speaker, a microphone, a buzzer, a touch panel, a switch, keys, a lamp, and the like. The HMI **314** is an example of an "internal notification device." For example, the HMI **314** notifies the occupant of the travel state of the vehicle **300** controlled by the control device **360** in a notification mode that varies with a difference in the travel state. The HMI **314**, for example, presents information from the control device **360** or presents information acquired from an external device with the communication device **312**.

The movement mechanism **320** is a mechanism for moving the vehicle **300** on a road. The movement mechanism **320** is, for example, a wheel group including a steered wheel and a drive wheel. Also, the movement mechanism **320** may be a leg for multi-legged walking.

The drive device **330** outputs a force to the movement mechanism **320** and moves the vehicle **300**. For example, the drive device **330** includes a motor for driving the drive wheel, a battery for storing electric power supplied to the motor, a steering device for adjusting a steering angle of the steered wheel, and the like. The drive device **330** may include an internal combustion engine, a fuel cell, or the like as a driving force output means or a power generation means. Also, the drive device **330** may further include a brake device due to friction force or air resistance. The drive device **330** may control the traveling of the vehicle **300** on the basis of information about steering control and speed control from the area monitoring server **400** instead of (or in addition to) operation content (a manual driving operation) of the operation element **306**.

The external notification device **340** is, for example, a lamp, a display device, a speaker, or the like provided on an outer panel of the vehicle **300** and configured to provide an information notification to the outside of the vehicle **300**. For example, the external notification device **340** provides a notification of the travel state of the vehicle **300** controlled by the control device **360** in the vicinity of the mobile object (within a prescribed distance from the vehicle **300**) in a notification mode that varies with a difference in the travel state.

FIG. **5** is a perspective view of the vehicle **300** seen from above. In FIG. **5**, FW denotes a steered wheel, RW denotes a drive wheel, SD denotes a steering device, MT denotes a motor, and BT denotes a battery. The steering device SD, the motor MT, and the battery BT are included in the drive device **330**. Also, AP denotes an accelerator pedal, BP denotes a brake pedal, WH denotes a steering wheel, SP denotes a speaker, and MC denotes a microphone. The vehicle **300** shown in FIG. **5** is a single-seater four-wheeled vehicle and an occupant P sits in a driver's seat DS and wears a seat belt SB. An arrow $\alpha 1$ is a travel direction (speed vector) of the vehicle **300**.

The external world detection device **302** is provided near the front end of the vehicle **300** and the internal camera **308** is provided at a position where the head of the occupant P can be imaged from the front of the occupant P. The external notification device **340** serving as a display device is provided near the front end of the vehicle **300**. In front of the occupant P inside of the mobile object, the HMI **314** serving as the display device is provided. The external notification device **340** may be formed integrally with a speaker SP and the HMI **314** may be formed integrally with the speaker SP and a microphone MC.

Returning to FIG. **4**, the storage device **350** is a non-transitory storage device such as, for example, a hard disk drive (HDD), a flash memory, or a random-access memory (RAM). The storage device **350** stores map information **352**, a program **354** executed by the control device **360**, and the like. The map information **352** includes, for example, road information associated with position information (a road shape (a width, curvature, and a slope) and positions of a stop line and a pedestrian crossing), point of interest (POI) information, traffic regulation information, address information (an address/postal code), facility information, telephone number information, and the like. The position information includes latitude and longitude. Although the storage device

350 outside of the frame of the control device 360 is described in FIG. 4, the storage device 350 may be included in the control device 360.

[Control Device]

The control device 360 includes, for example, a physical object recognizer 362 and a controller 364. For example, the physical object recognizer 362 and the controller 364 are implemented by a hardware processor such as a central processing unit (CPU) executing the program (software) 354. Also, some or all of these components may be implemented by hardware (including a circuit; circuitry) such as a large-scale integration (LSI) circuit, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or a graphics processing unit (GPU) or may be implemented by software and hardware in cooperation. The program may be pre-stored in the storage device 350 or may be stored in a removable storage medium (a non-transitory storage medium) such as a DVD or a CD-ROM and installed in the storage device 350 when the storage medium is mounted in a drive device.

The physical object recognizer 362 recognizes a surrounding situation of the vehicle 300 on the basis of an output of the external world detection device 302. For example, the physical object recognizer 362 recognizes a physical object located within a prescribed distance from the vehicle 300. Physical objects include some or all of mobile objects such as a vehicle, a bicycle, and a pedestrian, runway boundaries such as a road marking, a step, a guardrail, a shoulder, and a median strip, structures installed on the road such as a road sign and a signboard, and an obstacle such as a falling object that exists (falls) on the runway. For example, the physical object recognizer 362 acquires information of the presence, position, type, and the like of another mobile object by inputting an image captured by the external camera to a trained model trained to output the information of the presence, position, and type of a physical object when the image captured by the external camera of the external world detection device 302 is input. Other types of mobile objects can also be estimated on the basis of a size in the image, an intensity of a reflected wave received by a radar device of the external world detection device 302, and the like. Also, the physical object recognizer 362 may recognize the speed of another mobile object detected by the radar device using, for example, a Doppler shift or the like.

The physical object recognizer 362 recognizes a marking for dividing the road on which the vehicle 300 is traveling. For example, the physical object recognizer 362 recognizes the marking by, for example, analyzing the image captured by the external camera of the external world detection device 302. For the recognition of the marking, an output such as a radar device, a LIDAR sensor, or a sensor fusion device may be used as an auxiliary.

The physical object recognizer 362 may collate the position information of the vehicle 300 from the positioning device 310 with the map information 352 to recognize whether the vehicle 300 is traveling on the road (whether or not the vehicle 300 deviates from the road).

The controller 364 controls an overall configuration of the vehicle 300. For example, the controller 364 outputs information for providing a notification from the HMI 314 to the occupant on the basis of the information recognized by the physical object recognizer 362 or causes the external notification device 340 to output the information. The controller 364 causes the communication device 312 to transmit vehicle data including information acquired by the external world detection device 302, a surrounding situation recognized by the physical object recognizer 362, position infor-

mation of the vehicle 300 positioned by the positioning device 310, and a date and time information (an acquisition date and time, a recognition date and time, and a positioning date and time) to the area monitoring server 400. On the basis of the information provided from the area monitoring server 400, the controller 364 causes the HMI 314 to output information, causes the external notification device 340 to output information, or causes the drive device 330 to execute travel control.

[Area Monitoring Server]

FIG. 6 is a diagram showing an example of a configuration of the area monitoring server 400. The area monitoring server 400 includes, for example, a server-side communicator 410, an acquirer 420, an analyzer 430, a manager 440, a provider 450, and a server-side storage 460. The acquirer 420, the analyzer 430, the manager 440, and the provider 450, for example, are implemented by a hardware processor such as a CPU executing the program (software). Also, some or all of these components may be implemented by hardware (including a circuit; circuitry) such as an LSI circuit, an ASIC, an FPGA, or a GPU or may be implemented by software and hardware in cooperation. The program may be pre-stored in the storage device or may be stored in a removable storage medium (a non-transitory storage medium) such as a DVD or a CD-ROM and installed in the storage device when the storage medium is mounted in a drive device.

The server-side storage 460 may be implemented by the above-described various types of storage devices, a solid-state drive (SSD), an electrically erasable programmable read-only memory (EEPROM), a ROM, a random-access memory (RAM), or the like. The server-side storage 460 stores, for example, a monitoring information database (DB) 462, infrastructure facility information 464, map information 466, programs, other various types of information, and the like.

The monitoring information DB 462 stores content such as sensor data transmitted from the sensor device 100 and vehicle data transmitted from the vehicle 300. In the infrastructure facility information 464, for example, an installation position, an installation orientation (an angle of view (an imaging range) or a display direction) and type information are associated with a sensor ID of the sensor device 100 or identification information (an information providing device ID) of the information providing device 200. For example, in the case of the sensor device 100, the type information includes identification information for identifying a stereo camera, a monocular camera, a fisheye camera, an infrared camera, a TOF camera, a radar device, a LIDAR sensor, a microphone, and the like. In the case of the information providing device 200, the type information includes, for example, identification information for identifying a display, a speaker, and a light emitter. The map information 466 includes information similar to the map information 352. The map information 466 may be updated at any time by the server-side communicator 410 communicating with an external device.

The server-side communicator 410 communicates with the sensor device 100, the information providing device 200, the vehicle 300, and other external devices via the network NW.

The acquirer 420 acquires information from the sensor device 100, the vehicle 300, and other external devices. For example, the acquirer 420 acquires the sensor data from the sensor device 100 or the vehicle data from the vehicle 300, and stores the acquired data in a monitoring information DB 472.

The analyzer **430** analyzes sensor data and vehicle data acquired by the acquirer **420**. For example, the analyzer **430** analyzes image data when the sensor data or the vehicle data includes the image data. For example, the analyzer **430** recognizes a surrounding situation of the sensor device **100** by converting the image data into coordinates of an overhead view coordinate system and performing an image analysis process in a well-known method (a binarization process, a contour extraction process, an image enhancement process, a feature-quantity extraction process, a pattern matching process, or the like) on the basis of the coordinates after the conversion. The analyzer **430** may perform the above-described image analysis process without performing coordinate conversion.

For example, the analyzer **430** identifies a physical object included in the image data on the basis of a matching result (a matching degree) between feature information obtained by analyzing the image data and feature information predetermined for each physical object. The matching degree may be derived, for example, according to the number of matching elements among a plurality of feature elements included in the feature information and may be derived according to a degree of similarity of each feature element or the entire feature information. The matching degree may be derived, for example, on the basis of a sum of differences for each element. The matching degree may be derived from two items of information to be compared using functions based on artificial intelligence (AI) such as machine learning (a neural network) and deep learning or may be derived using other methods.

The analyzer **430** may recognize a position, type, speed, and the like of a physical object located in the monitoring target area included in the image data. Examples of the physical object include a mobile object such as the vehicle **300**, a pedestrian, or a bicycle. The physical object may include a road structure or the like. Road structures include, for example, a road sign, a traffic signal, a curb, a median, a guardrail, a fence, a wall, a railroad crossing, a pedestrian crossing drawn on the road surface, a stop line, and the like. The physical object may include an obstacle that obstructs (or is likely to obstruct) the traveling of the vehicle **300**.

The analyzer **430** recognizes a position (a relative position) of the mobile object located in the monitoring target area or recognizes a state of the mobile object such as a speed, acceleration, or movement direction using time-series image data. The position of the physical object is recognized, for example, as the position of an absolute coordinate system with the origin point of the representative point of the mobile object (the center of gravity, the center of a drive axis, or the like). The “state” of the physical object may include, for example, acceleration or jerks of the mobile object, or an “action state” (for example, whether or not it is passing or attempting to cross a pedestrian crossing). The analyzer **430** may recognize a physical object in a process similar to the physical object recognition process performed by the physical object recognizer **362**.

The analyzer **430** may analyze feature information such as a shape for identifying a mobile object included in the monitoring target area. The feature information may include a pattern (including symbols, numbers, and the like visible from the outside), a color, and the like in addition to the shape. The analyzer **430** may measure a travel period of time of the mobile object in a prescribed segment.

The analyzer **430** may analyze a difference between a position of the road included in the image data and a position of the road included in reference image data (a reference image) obtained by performing an imaging process in

advance by the same sensor device **100**. The reference image is an image taken at a correct position, direction, and angle of view (an imaging range) using the same sensor device **100**. For example, a degree of deviation between the road situation based on the image data and the road situation based on a predetermined reference image is derived. The degree of deviation is an index value indicating a magnitude of the deviation and the degree increases as the deviation increases. For example, the analyzer **430** binarizes the captured image data and the reference image data and analyzes the degree of deviation from an amount of deviation in a distance or direction of a road marking or the like obtained from a binarization result.

The analyzer **430** may extract an image area used as an analysis target among all image areas of the image data on the basis of a season, weather, or time period in which the image data is acquired and analyze the amount of deviation using an image of an extracted area (an extracted image). Thereby, it is possible to suppress the misrecognition of a physical object included in the image by moving the shadow by a building, tree, or the like around the monitoring target area.

When a mobile object is moving outside of a sensor range of a plurality of sensor devices in the monitoring target area, the analyzer **430** may monitor the situation of the monitoring target area using vehicle data obtained by sensors (external world detection devices **302**) mounted in other mobile objects. The analyzer **430** may recognize information of at least one item from information about the movement of the mobile object from the sensor data and a reflection intensity corresponding to an attribute of the mobile object. The information about the movement of the mobile object, for example, includes various types of information that change with movement (in which an amount of movement can be derived) such as a deviation in the position of the mobile object and a speed distribution. The deviation in the position of the mobile object is, for example, a deviation in the position in a horizontal direction (a road width direction) for an extension direction (a longitudinal direction) of the road. The deviation includes information such as proximity to the right than the center of the lane and information such as an amount of lateral position change in a prescribed period of time. The speed distribution is a speed pattern of the time series of the mobile object (for example, gradually decelerating or accelerating, or traveling at a constant speed). The attribute of the mobile object is, for example, a type of vehicle, pedestrian, bicycle, or the like.

The manager **440** manages a situation of the monitoring target area on the basis of an analysis result of the analyzer **430** and the like. For example, the manager **440** manages whether or not the vehicle **300** traveling on the road included in the monitoring target area tends to deviate from the road, whether or not the vehicle **300** is slipping, whether or not there is a possibility of contact with other physical objects, and the like. When a difference between a travel period of time of a prescribed segment of the vehicle **300** measured by the analyzer **430** and a predetermined standard travel period of time is greater than or equal to a threshold value, the manager **440** may determine that there is an abnormality in the vehicle **300** (for example, a poor physical condition of the occupant, a failure of the vehicle) or an abnormality in the monitoring target area (for example, road surface freezing).

The manager **440** may include, for example, a determiner **442**. The determiner **442** determines whether or not mobile objects detected by the plurality of sensor devices **100** are the same physical object on the basis of detection results of

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the plurality of sensor devices **100** installed in the vicinity of the monitoring target area. When it is determined that the mobile objects are the same physical object, the manager **440** tracks the behavior of the mobile object by integrating the analysis results of the sensors and the manager **440** tracks the behavior of the mobile object individually when it is determined that the mobile objects are not the same physical object. The determiner **442** may determine whether or not the degree of deviation from the reference image analyzed by the analyzer **430** is greater than or equal to the threshold value. When it is determined that the degree of deviation is greater than or equal to the threshold value, the manager **440** determines that maintenance of at least one of a peripheral monitoring area and the sensor device **100** is necessary. For example, the manager **440** determines that maintenance of the sensor device **100** is necessary when a difference between the position of the road included in the image captured by the sensor device **100** and the position of the road included in the reference image is greater than or equal to a threshold value.

The provider **450** provides information to a mobile object (a traffic participant such as the vehicle **300** or a pedestrian) moving in the monitoring target area via the vehicle **300** traveling in the monitoring target area, the sensor device **100** installed near the area, and the information providing device **200**. The information provided in this case includes, for example, an action instruction for the mobile object (for example, deceleration, stop, or the like). The provider **450** may provide information to a manager (for example, a maintenance staff member) or the like who manages the monitoring target area. In this case, the provided information includes maintenance locations and maintenance content (for example, an adjustment of the installation position of the sensor device or cleaning of the monitoring target area). [Specific Example of Area Monitoring]

Next, specific examples of an area monitoring process of the area monitoring system **1** will be described separately as several examples.

<First Monitoring Example>

FIG. **7** is a diagram for describing a first monitoring example. The first monitoring example includes, for example, integrating information obtained from a plurality of sensor devices **100** installed on a road (an example of a monitoring target area) to comprehensively monitor a situation on a monitoring target road. In the example of FIG. **7**, vehicles **300-1** and **300-2** traveling in a lane **L1** defined by road markings **LR** and **LL** and sensor devices **100-1** and **100-2** installed near the lane **L1** are shown. The sensor device **100-1** is an example of a “first sensor device,” and the sensor device **100-2** is an example of a “second sensor device.” It is assumed that the vehicle **300-1** is traveling at a speed **V1** in the extension direction of lane **L1** and the vehicle **300-2** is traveling behind the vehicle **300-1** at a speed **V2** in a direction identical to that of the vehicle **300-1**. The sensor devices **100-1** and **100-2** are installed at prescribed intervals and have fixed angles of view (imaging ranges) **AR1** and **AR2**, respectively, to image an area including at least apart of the road (the lane **L1**). Sensor data including the captured camera image is transmitted to the area monitoring server **400** through the communicator **110**. The installation positions, directions, and angles of view of the sensor devices **100-1** and **100-2** (information about a reference image) are registered in the infrastructure facility information **464** in advance.

The acquirer **420** of the area monitoring server **400** acquires sensor data transmitted from the sensor devices **100-1** and **100-2**. The acquired sensor data may be stored in

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the monitoring information DB **462**. The analyzer **430** analyzes the sensor data and recognizes the positions of the vehicles **300-1** and **300-2**, the positions of the road markings **LR** and **LL**, and the like. The analyzer **430** analyzes speeds of the vehicles **300-1** and **300-2** from an amount of movement (an amount of change in position) in a prescribed period of time obtained from time-series sensor data. The analyzer **430** identifies vehicles included in sensor data acquired from the plurality of sensor devices **100** on the basis of a matching degree of feature information such as a shape of the vehicle **300** (for example, a roof portion of the vehicle **300**). For example, the analyzer **430** identifies a vehicle having the matching degree greater than or equal to a threshold value as the same vehicle (the same physical object), assigns common identification information to the vehicle, and assigns other different identification information to a vehicle whose matching degree is less than the threshold value. Thereby, the manager **440** can manage a movement situation for each vehicle on the basis of the assigned identification information. The analyzer **430** may perform a similar analysis process on mobile objects other than the vehicle **300** located in the monitoring target area.

The manager **440** tracks a travel situation of the vehicle **300** on the basis of an analysis result and manages whether or not the vehicles **300-1** and **300-2** tend to depart from the lane **L1** or manages whether or not the behavior of the vehicles **300-1** and **300-2** is abnormal due to slips or the like. For example, the determiner **442** of the manager **440** acquires a movement speed and a movement direction of the vehicle **300** between prescribed points according to an analysis result and determines that the vehicle **300** tends to depart from the lane **L1** when it is determined that the vehicle **300** is likely to step over the road markings **LR** and **LL** within a prescribed period of time from positional relationships between the acquired movement speed or the acquired movement direction and the road markings **LR** and **LL**.

The manager **440** determines that the vehicle **300** is slipping when an amount of change in an angle formed by the extension direction of the lane **L1** (the road) and the travel direction of the vehicle **300** in a prescribed period of time (a rate of change in the yaw angle (a yaw rate (an angular speed)) for the travel direction within a prescribed period of time) is greater than or equal to a threshold value.

Here, the plurality of sensor devices **100** installed in the monitoring target area are continuously installed at intervals so that sensor ranges (for example, angles of view of the camera devices) overlap and therefore the same vehicle or the same person can be tracked more accurately. However, in practice, due to the influence of the road shape, facility cost, or the like, the sensor ranges may be installed without overlapping, in which case a blind spot occurs in the monitoring target area. Therefore, in the first monitoring example, in a situation where the sensor ranges do not overlap, when it is estimated that a passing vehicle is outside of the sensor range of one of the plurality of sensor devices **100** and heading into the sensor range of another sensor device **100**, information of a segment where the sensor ranges do not overlap is acquired from a detection result of the external world detection device **302** mounted in the vehicle **300**.

For example, when the travel situation of the vehicle **300-1** shown in FIG. **7** is monitored, the behavior of the vehicle **300-1** outside of the sensor ranges (the angles of view **AR1** and **AR2**) of the sensor devices **100-1** and **100-2** is tracked using image data obtained in the imaging process of the external world detection device **302** mounted in the

vehicle **300-2**. Thereby, lane departure, slip, and the like of the vehicle **300-1** in the blind spot area not included in the angles of view **AR1** and **AR2** can be recognized.

In the first monitoring example, even if the sensor ranges of the plurality of sensor devices **100** overlap, a tracking process may be performed using a result of analyzing image data obtained in an imaging process of the external world detection device **302** mounted in the vehicle **300-2** similarly even in a case where the blind spot area has occurred by performing an analysis process using an image extracted from the entire image area of the image data on the basis of a season, weather, time period, or the like.

Thus, according to the first monitoring example, the blind spot area can be reduced by interpolating the blind spot of the sensor data from the sensor device **100** using the vehicle data acquired from the vehicle **300-2** and a situation occurring in the monitoring target area can be monitored without omission.

<Second Monitoring Example>

FIG. **8** is a diagram for describing a second monitoring example. The second monitoring example includes, for example, monitoring the behavior of the vehicle **300** on the basis of a result of analyzing images captured by the plurality of sensor devices **100** installed in the monitoring target area and a road shape included in the monitoring target area. In the example of FIG. **8**, a road of a T-junction in which the lane **L2** is connected perpendicular to the lane **L1** is shown.

In the second monitoring example, the sensor devices **100-1** and **100-2** are installed at positions where the sensor ranges (angles of view **AR1** and **AR2**) do not overlap. The manager **440** tracks the travel situation of the vehicle **300-1** on the basis of the sensor devices **100-1** and **100-2**. Here, when it is estimated that the vehicle **300** located at the angle of view **AR1** of the sensor device **100-1** moves outside of the angle of view **AR1** and heads in the direction of the angle of view **AR2** of the sensor device **100-2**, the manager **440** predicts that an abnormality is likely to have occurred in the vehicle **300** or that the vehicle **300** is stopped when the vehicle **300** is not recognized in the sensor data captured by the sensor device **100-2** within a prescribed period of time. The manager **440** acquires a road shape around the installation position of the sensor device **100** with reference to the map information **466** on the basis of the position information of the sensor devices **100-1** and **100-2** and predicts that the vehicle **300** is likely to move to a lane **L2** when the lane (branching lane) **L2** connected to the lane **L1** is located between the sensor ranges of the sensor devices **100-1** and **100-2**. The lane **L2** is an example of a "branching road."

The manager **440** may predict that an abnormality such as an accident is likely to occur in a segment between the angles of view **AR1** and **AR2** when the vehicle **300** is not recognized in the sensor data obtained in the imaging process of the sensor device **100-2** within a prescribed period of time and when other vehicles are not recognized (or the number of passing vehicles is less than a threshold value). According to the first and second monitoring examples, the monitoring target area can be more appropriately monitored using a plurality of sensor devices **100** installed on the road.

<Third Monitoring Example>

FIG. **9** is a diagram for describing a third monitoring example. The third monitoring example includes monitoring an installation situation of the sensor device **100** installed near the monitoring target area. In the example of FIG. **9**, a vehicle **300** traveling in an X-axis direction in FIG. **9** on a road **RD1** where oncoming vehicles can pass is shown. For

example, the direction and sensor range of the sensor device **100** may deviate from the standard due to an influence of an earthquake, wind, rain, or the like. If the deviation is not detected at an early stage, there is a possibility that it will be impossible to recognize the physical object or that accurate area monitoring will be impossible due to erroneous recognition or the like. Therefore, in the third monitoring example, the manager **440** aggregates a position (a travel road position) where the vehicle **300** traveling in the same direction (for example, the X-axis direction) passes on the road **RD1** in the monitoring target area and monitors an installation situation of the sensor device **100** on the basis of an aggregation result.

For example, the manager **440** acquires a difference (a pixel deviation amount) **W1** between an actually recognized traveling path position obtained as an aggregation result and a predetermined travel path position for the sensor device **100** installed on the road and determines that there is an abnormality in the state (the installation state) of the sensor device **100** is abnormal when the acquired difference **W1** is greater than or equal to a threshold value.

Although the determination is made on the basis of a deviation amount **W1** in a lateral direction of the road **RD1** in the example of FIG. **9**, a deviation amount of each of a longitudinal direction, an oblique direction, a rotation direction, and the like of the road **RD1** may be acquired instead of (or in addition to) this and it may be determined whether or not there is an abnormality in the state of the sensor device **100** on the basis of an acquired multidimensional deviation amount.

When it is determined that the total number of pixels recognized as a road area from the image data is less than a threshold value, the manager **440** may predict that fallen leaves, other obstacles, and the like are likely to be located on the road.

When it is predicted that shadows will appear on the road within the monitoring target area due to a direction of solar radiation, a surrounding environment (for example, trees, buildings, and the like), and external lights installed in the monitoring target area, the total number of pixels recognized as those of a road is likely to change. Therefore, the manager **440** may adjust the total number of pixels for each time period, set an area in which a recognition process is not performed within the image in consideration of the position and range of the shadow in the time period, and monitor a situation of the road based on the total number of pixels of the road using an image outside of the area. The manager **440** may adjust the time period for each season and may determine whether or not to perform the above-described process according to the presence or absence of a shadow due to weather.

In the third monitoring example, the manager **440** may determine whether or not maintenance regarding at least one of the monitoring target area and the sensor device **100** is necessary on the basis of the monitoring result. For example, the maintenance includes, for example, the maintenance of a road included in the monitoring target area and the maintenance of the sensor device **100** installed in the monitoring target area. The maintenance of the road includes, for example, cleaning up fallen leaves and garbage on roads, removing obstacles, draining puddles, and the like. The maintenance of the sensor device **100** includes, for example, correcting (readjusting) a deviation in the installation direction of the sensor device **100**, repairing or replacing the device itself, and the like. Here, specific examples in which

the maintenance of the road is monitored and the maintenance of the sensor device **100** is monitored will be described below.

<Monitoring for Maintenance of Road>

In the case of monitoring for the maintenance of the road, the manager **440** determines that maintenance of the road included in the camera image is necessary, for example, when a first deviation amount between the reference image and the camera image is greater than or equal to a prescribed amount and causes the provider **450** to provide information for promoting maintenance. The first deviation amount is, for example, a difference value in the number of pixels (the number of bits) when the total number of pixels in a part recognized as a road of the camera image is compared with that in the same part of the reference image. For example, when a camera image of a road where fallen leaves or the like are accumulated on the road surface is binarized, an area where fallen leaves are accumulated is not recognized as a road and the number of pixels in the part recognized as the road in the camera image is likely to be smaller than that in the reference image. When there is a puddle on or around the road or the like, the number of pixels is likely to be larger than that of the reference image according to an imaging environment. Therefore, the manager **440** determines that the maintenance of the road is necessary when a degree of deviation based on the first deviation amount is greater than or equal to a threshold value (i.e., when the deviation is large).

<Monitoring for Maintenance of Sensor Device **100**>

In the case of monitoring for the maintenance of the sensor device **100** (for example, the camera device), for example, the manager **440** causes the provider **450** to provide information for promoting the maintenance of the sensor device **100** when a difference between the position of the road included in the reference image and the position of the road included in the camera image is greater than or equal to a threshold value. For example, when a second deviation amount between the reference image and the camera image is greater than or equal to a prescribed amount, the manager **440** determines that it is necessary to maintain the camera device capturing the camera image, and causes the provider **450** to provide information for promoting the maintenance. The second deviation amount is a difference value between an output result obtained by multiplying parts recognized (set) as those of the road in the binary reference image and the binary camera image and outputting multiplication results and the number of pixels (the number of bits) recognized as those of the road in the binary reference image. The term "multiplying the parts by each other and outputting the multiplication results" indicates, for example, setting portions (pixels) recognized as part of the road in each binary image to "1," setting other portions (pixels) to "0," multiplying values (1 or 0) of pixels at the same position in the images by each other, and outputting a sum obtained by summing multiplication results for all pixels. The manager **440** determines that maintenance of the sensor device **100** is necessary when the degree of deviation based on the second deviation amount is greater than or equal to the threshold value (i.e., when the deviation is large). The first deviation amount and the second deviation amount described above are analyzed, for example, by the analyzer **430**.

Thus, according to the third monitoring example, an abnormality in the position of the sensor device **100** and an abnormality in the road can be detected at an early stage, and a maintenance staff member or the like can be notified of the abnormality.

<Fourth Monitoring Example>

Next, a fourth monitoring example will be described. The fourth monitoring example includes monitoring the monitoring target area by combining sensor data obtained from a plurality of sensor devices having different types (characteristics). FIG. **10** is a diagram showing an example of characteristic information for the sensor type. In the example of FIG. **10**, for each sensor type, evaluation results for position accuracy, speed accuracy, a detection range, situation ascertainment, and environmental resistance are shown as the characteristic information. In the example of FIG. **10**, it is assumed that the mark "⊙" indicates the highest evaluation, and the marks "○" and "Δ" indicate sequentially lower evaluations. In FIG. **10**, evaluation results may be included for each sensor type in terms of cost.

For example, in a segment having a prescribed distance such as a road, it is necessary to detect a physical object within the monitoring target area using a plurality of sensors. Because the plurality of sensor devices may not have the same characteristics (performance and functions), an appropriate combination according to the monitoring target area is necessary. Thus, in the fourth monitoring example, on the basis of the characteristics of each sensor type determined in advance as shown in FIG. **10**, the types and number of sensors optimal for the monitoring target area are installed in consideration of characteristics of each sensor and the situation of the monitoring target area is monitored. For example, a stereo camera, a TOF camera, a LIDAR sensor, or the like is used in an area where a position of a physical object is to be more accurately monitored and a radar device is installed in an area where the speed of the physical object is more accurately monitored. A radio wave sensor may be installed in a segment where the curvature of the road included in the monitoring target area is less than a threshold value and an optical sensor may be installed in a segment where the curvature of the road is greater than or equal to the threshold value. The manager **440** manages the types (combinations) and number of sensor devices **100** installed in the vicinity of the monitoring target area on the basis of the characteristic information, road shape, and the like as shown in FIG. **10**. The manager **440** may adjust the types and number of sensors to be installed in accordance with the cost of the sensor device **100**.

For example, when a radio wave sensor and an optical sensor are installed in the monitoring target area, a mobile object (for example, the vehicle **300**) is recognized by the optical sensor and then moves outside the sensor range, and the radio wave sensor is located at a movement destination, the determiner **442** determines whether or not physical objects detected by the sensors are the same physical object on the basis of at least one information item of information about the movement of the mobile object (a position deviation and a speed distribution) and the reflection intensity according to the attribute of the mobile object capable of being acquired from the sensor data. For example, the determiner **442** determines whether or not physical objects detected by the sensors are the same physical object by selecting at least one information item among the deviation in the position of the mobile object, the speed distribution, and the reflection intensity according to the sensor types of the first sensor device and the second sensor device.

The determiner **442** may determine whether or not the detected physical objects are the same physical object using similar information even if the mobile object moves outside the sensor range after the mobile object is recognized by the radio wave sensor and the optical sensor is located at the movement destination.

Further, the manager **440** may issue an action instruction to the vehicle **300** when the above-described mobile object is a vehicle **300** and a physical object (for example, another vehicle, a pedestrian, or a bicycle) is located in the vicinity of the vehicle **300** (in other words, when there is a physical object that is likely to come into contact with the vehicle **300**). Action instructions include, for example, instructions regarding one or both of speed control (deceleration and stop) and steering control.

The manager **440** may differentiate the content of the action instruction to the vehicle **300** according to, for example, a degree of influence on the contact of a physical object approaching the vehicle **300**. For example, if the approaching physical object is a traffic participant (a traffic participant with a high degree of influence on the body such as injury) having little resistance to the vehicle **300** such as a person or a bicycle, action content is strengthened (a large-action instruction is issued) as compared with the case where the approaching physical object is another vehicle. Specifically, when a person is approaching the vehicle **300** (when the person is within a prescribed distance), a stronger deceleration instruction is issued than when the vehicle **300** approaches. Thereby, safety can be further improved.

When the information providing device **200** is set in the monitoring target area, the provider **450** causes the information providing device **200** to output an image or sound indicating the content of the action instruction under the control of the manager **440**. Thereby, the occupant of the vehicle **300** can perform driving to avoid contact with a physical object according to the action instruction output from the information providing device **200**. When a plurality of sensor devices **100** include a camera device (an image sensor), the manager **440** may acquire information about the attribute of a physical object approaching the vehicle **300** included in the camera image of the camera device and may control the provider **450** so that the acquired information is output to the information providing device **200**. Thereby, because the information providing device **200** installed near the approaching physical object can cause the attributes of the approaching physical object (a person, a vehicle, a bicycle, or the like) as well as the action instruction to be displayed, it is possible to allow a target physical object associated with the action instruction to be accurately ascertained.

The provider **450** may output information for causing the vehicle **300** to execute one or both of the speed control and steering control according to content of the action instruction for the vehicle **300** to the vehicle **300** under the control of the manager **440**. Thereby, travel control of the vehicle **300** can be executed without waiting for the driving operation of the occupant of the vehicle **300**.

Thus, according to the fourth monitoring example, the situation can be more appropriately monitored using sensor data even if different types of sensor devices **100** are installed in the monitoring target area. According to the fourth monitoring example, because lower-cost sensor devices can be combined in accordance with monitoring content and the like, the facility cost can be reduced. Each of the first to fourth monitoring examples described above may include some or all of the other monitoring examples. [Information Provision]

Next, an example of information provision based on the monitoring result will be described with reference to the drawings. FIG. **11** is a diagram showing a first example of information provision. In the example of FIG. **11**, for example, when it is determined that the vehicle **300** departs from a lane **L1** on the basis of the first monitoring example,

information is provided to the occupants of the vehicle **300** so that movement is possible in a non-deviation direction. In the example of FIG. **11**, the sensor device **100** is installed on a curve path (a segment where the curvature is greater than a prescribed level) from which the vehicle is likely to depart. The information providing device **200** is installed near the lane **L1**.

The manager **440** tracks the traveling of the vehicle **300** on the basis of the sensor data obtained from the sensor device **100**. Here, when it is determined that the travel direction of the vehicle **300** (an arrow **A1** in FIG. **11**) may depart from an area of the lane **L1** (step over a road marking **LR**) on the basis of a tracking result, the provider **450** causes the information providing device **200** to display information (action instructions) for suppressing the departure. In the example of FIG. **11**, text information such as "Please steer to the left" is displayed on the display **220** of the information providing device **200** installed at a position where the occupants of the vehicle **300** can see. The manager **440** manages which information providing device **200** displays this text information.

The provider **450** may transmit information (an action instruction) for causing the occupant of the vehicle **300** to execute the steering operation to the vehicle **300** via the sensor device **100** and cause the HMI **314** to display the information (the action instruction). Alternatively, the provider **450** may transmit control information for automatically performing steering control of the vehicle **300** to the vehicle **300** and cause the drive device **330** to execute a travel control process of steering the vehicle **300** to the left. Thereby, the vehicle **300** can be moved in a direction of an arrow **A2** in FIG. **11**.

When the manager **440** determines that the vehicle **300** is slipping, the provider **450** may cause the display **220** of the information providing device **200** to display caution information such as "road surface freezing caution" or may transmit information (the action instruction) for allowing the occupant of the vehicle **300** to execute a deceleration operation (or automated deceleration control) to the vehicle **300**.

Although an example in which information from the area monitoring server **400** is transmitted to the information providing device **200** or the vehicle **300** via the sensor device **100** is shown in the example of FIG. **11**, the information may be directly transmitted from the area monitoring server **400** to the information providing device **200** or the vehicle **300**.

FIG. **12** is a diagram showing a second example of information provision. In the example of FIG. **12**, a notification is given to avoid contact between mobile objects moving in a lane **L1**. In the example of FIG. **12**, a vehicle **300**, a pedestrian **P1**, and a bicycle **P2** moving in the lane **L1** are shown. In the vicinity of the lane **L1** shown in FIG. **12**, the sensor device **100** configured to image an area including the lane **L1** and the information providing device **200** are installed. Arrows **A3**, **A4**, and **A5** shown in FIG. **12** indicate movement directions of the vehicle **300**, the bicycle **P2**, and the pedestrian **P1** acquired from the time-series sensor data.

The manager **440** of the area monitoring server **400** analyzes the sensor data detected by the sensor device **100**, determines which of the vehicle **300**, the pedestrian **P1**, and the bicycle **P2** is likely to come into contact with another physical object from an analysis result, and provides information (an action instruction) for avoiding contact to the sensor device **100**, the information providing device **200**, the vehicle **300**, and the like when it is determined that there is a possibility of contact.

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For example, the provider 450 transmits provided information for displaying a text image such as "Please be careful of contact" from the display 220 of the information providing device 200 to the information providing device 200 via the network NW. Thereby, the occupant of the vehicle 300, the pedestrian P1, and the occupant of the bicycle P2 seeing text displayed on the display 220 of the information providing device 200 can avoid contact as soon as possible.

The provider 450 may transmit information to the vehicle 300 so that information similar to that displayed on the display 220 is displayed on the HMI 314 of the vehicle 300 or may transmit information for notifying that the pedestrian P1 or the bicycle P2 is approaching. The provider 450 may transmit control information for stopping the vehicle 300 to the vehicle 300 or may transmit instruction information for outputting a warning sound (horn) for notifying the pedestrian P1 and the occupant of the bicycle P2 that the vehicle 300 is approaching or the like from the external notification device 340 of the vehicle 300 to the vehicle 300. Thereby, more appropriate information can be provided in accordance with the situation of the physical object passing on the road (in the lane L1), and safety during passing the lane L1 can be further improved.

When information is provided in the third monitoring example, the manager 440 may provide information requesting maintenance (environmental maintenance) to a terminal device (not shown) of the maintenance staff member. In this case, the notification of a position of the sensor device 100 where an abnormality has occurred, a position of the monitoring target area, and a type of abnormality (for example, movement of the sensor device 100) may be provided. Thereby, the maintenance staff member can more specifically ascertain a place and content of maintenance and the like and can perform more appropriate work preparation and work.

Furthermore, when a light emitter is provided in the information providing device 200, the light emitter may be controlled so that it is in a light emitting state in accordance with the action instruction. For example, when it is determined that the vehicle 300 tends to depart from the lane L1 as shown in FIG. 11, the provider 450 causes the light emitter installed in the road marking LR to be turned on in addition to (or instead of) the above-described information provision. As shown in FIG. 12, when mobile objects are close to each other, the light emitter installed on the road surface is turned on. The provider 450 may switch between lighting and blinking or adjust the color and light amount (intensity) in accordance with a distance between the vehicle 300 and the road marking LR and a degree of proximity (a relative distance) between the mobile objects. Thereby, it is possible to allow mobile object to more appropriately ascertain a current situation.

[Processing Sequence]

FIG. 13 is a sequence chart showing an example of a flow of a process executed by the area monitoring system 1 of the embodiment. In the example of FIG. 13, a process using the sensor device 100, the vehicle 300, the area monitoring server 400, and the information providing device 200 will be described. In the example of FIG. 13, it is assumed that the sensor device 100 is a camera device, the vehicle 300 is a vehicle traveling in an angle of view (a monitoring target area) at which an imaging process of the sensor device 100 is performed, and the information providing device 200 is installed near the monitoring target area of the sensor device 100.

In the example of FIG. 13, the sensor device 100 images the monitoring target area (step S100) and transmits sensor

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data including a captured camera image to the area monitoring server 400 (step S102). The vehicle 300 detects a surrounding situation of the monitoring target area with the external world detection device 302 or the like (step S104) and transmits vehicle data including a detection result to the area monitoring server 400 (step S106).

The area monitoring server 400 receives the sensor data and the vehicle data and analyzes the image data and the like included in the received data (step S108) and manages the monitoring target area on the basis of analysis results and the like (step S110). Specifically, at least one of the first to fourth monitoring examples described above is executed. Subsequently, when the area monitoring server 400 determines that it is necessary to provide information to the monitoring target area, the area monitoring server 400 generates the provided information (step S112) and transmits the generated provided information to the information providing device 200 (step S114). The area monitoring server 400 transmits the generated provided information to the vehicle 300 (step S116).

The information providing device 200 receives the information transmitted from the area monitoring server 400 and causes the display 220 or the like to display an image corresponding to the received information (step S118). Instead of (or in addition to) displaying an image on the display 220, the information providing device 200 may cause a sound corresponding to the received information to be output from the speaker 230. The light emitter provided on the road may be turned on (or blinked) in a state corresponding to the received information.

The vehicle 300 receives the information transmitted from the area monitoring server 400 and causes the HMI 314 to display an image corresponding to the received information, or executes travel control (for example, speed control or steering control) corresponding to the received information (step S120). Instead of (or in addition to) the above-described control, the vehicle 300 may output a sound corresponding to the received information to the external notification device 340. Thereby, the process of the present sequence ends.

In the processing of step S112, the area monitoring server 400 may generate information to be provided to the maintenance staff member when maintenance work of the monitoring target area is required and transmit the generated information to the terminal device owned by the maintenance staff member.

Subsequently, the specific processing of steps S108 to S112 described above will be described using a flowchart. <First Process>

FIG. 14 is a flowchart showing an example of a first process. In the example of FIG. 14, it is assumed that the sensor device 100 installed in the monitoring target area is a camera device and an example of a mobile object is the vehicle 300. The process shown in FIG. 14 may be iteratively executed at a prescribed interval or at a prescribed timing. It is assumed that these are also similar to those of a process shown in FIG. 15 to be described below (a second monitoring process).

In the example of FIG. 14, the acquirer 420 acquires camera data from the camera device (step S200). Subsequently, the analyzer 430 analyzes the acquired camera data (step S202). The manager 440 assigns identification information for each vehicle included in the camera data on the basis of a shape of the vehicle obtained as an analysis result (step S204) and manages a travel situation of the vehicle 300 in the monitoring target area (step S206).

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Subsequently, the manager 440 determines whether or not the vehicle 300 has departed from a lane (a road) during traveling or whether the vehicle 300 is slipping (step S208). When it is determined that the vehicle has departed or is slipping, the provider 450 generates information to be provided to the vehicle 300 (step S210) and causes the generated information to be output to the information providing device 200 (step S212). In the processing of step S212, the information to be provided to the vehicle 300 may be transmitted to the vehicle 300. In this case, the information to be provided to the vehicle 300 includes information to be output from the HMI 314 (an image or sound) or information for performing steering control and speed control. Thereby, the process of the present flowchart ends. If it is determined that there is no lane departure and slip of the vehicle 300 in the processing of step S208, the process of the present flowchart also ends. In the above-described processing of step S208, it may be determined whether or not mobile objects moving in the monitoring target area included in the camera data are likely to come into contact with each other. <Second Process>

FIG. 15 is a flowchart showing an example of a second process. In the example of FIG. 15, the acquirer 420 acquires camera data captured by the camera device installed in the monitoring target area (step S300). Subsequently, the analyzer 430 analyzes the acquired camera data (step S302). The manager 440 derives a degree of deviation between a road state based on camera data obtained as an analysis result and a road state based on a prescribed reference image (step S304). Subsequently, the determiner 442 determines whether or not the degree of deviation is greater than or equal to a threshold value (step S306). When it is determined that the degree of deviation is greater than or equal to the threshold value, the provider 450 generates information for promoting maintenance regarding at least one of the monitoring target area and the camera device (an example of the sensor device) (step S308) and causes the generated information to be output to the terminal device of the maintenance staff member, which is an example of the information providing device 200 (step S310). Thereby, the process of the present flowchart ends. In the processing of step S306, when it is determined that the degree of deviation is not greater than or equal to the threshold value, the process of the present flowchart ends. <Third Process>

FIG. 16 is a flowchart showing an example of a third process. In the example of FIG. 16, it is assumed that a plurality of sensor devices 100 installed in the monitoring target area are likely to include a radio wave sensor and an optical sensor. The process shown in FIG. 16 may be iteratively executed at a prescribed interval or at a prescribed timing. In the example of FIG. 16, the acquirer 420 acquires sensor data from the sensor device 100 (step S400). Subsequently, the analyzer 430 analyzes the acquired sensor data (step S402). Subsequently, the determiner 442 determines whether or not the sensor device 100, which has acquired the sensor data, includes sensor data of the radio wave sensor and the optical sensor. When it is determined that the sensor data of the radio wave sensor and the optical sensor is included, the determiner 442 determines the same physical object on the basis of information of at least one of deviations in positions of the mobile objects, speed distributions, and reflection intensities (step S406). The deviation and speed distribution of the position of the mobile object are examples and other information regarding the movement of the mobile object may be used. When it is determined that the sensor data of the radio wave sensor and the optical

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sensor is not included (only sensor data of one sensor), the determiner 442 determines the same physical object on the basis of a degree of similarity of the analysis results of the sensors (step S408). Subsequently, the manager 440 manages the movement state of the same physical object (step S410). Thereby, the present flowchart ends. In the third monitoring process, after step S410, slip, lane departure, and the like of the same physical object may be monitored and information may be provided on the basis of a monitoring result. Furthermore, it may be monitored whether or not it comes into contact with other physical objects and information based on a monitoring result may be provided.

Modified Examples

At least some of the components of the area monitoring server 400 described above may be provided in the sensor device 100, the information providing device 200, or the vehicle 300. For example, when the functions of the analyzer 430 are provided in the sensor device 100 or the vehicle 300, analysis results are transmitted to the area monitoring server 400.

At least some of the components of the information providing device 200 in the embodiment may be provided in the sensor device 100 and at least some of the components of the sensor device 100 may be provided in the information providing device 200.

According to the above-described embodiment, the area monitoring system 1 includes the plurality of sensor devices 100 each installed in a monitoring target area and configured to detect a mobile object moving in the monitoring target area; and the determiner 442 configured to determine whether or not mobile objects detected by the plurality of sensor devices 100 are the same physical object on the basis of detections results of the plurality of sensor devices 100, wherein the plurality of sensor devices 100 include at least a radio wave sensor and an optical sensor, and wherein the determiner 442 determines whether or not mobile objects detected by the radio wave sensor and the optical sensor are the same physical object on the basis of information of at least one of feature quantities regarding movements of the mobile objects and reflection intensities corresponding to attributes of the mobile objects when the mobile objects have been detected by the radio wave sensor and the optical sensor, whereby it is possible to more appropriately monitor the monitoring target area. Therefore, it will contribute to the development of a sustainable transportation system.

For example, according to the embodiment, a plurality of sensor devices 100 provided in the monitoring target area can be linked to more accurately track a mobile object and ascertain a situation. According to the embodiment, an infrastructure cooperative monitoring system can be provided by effectively utilizing an infrastructure facility such as cameras installed in a prescribed area. Because the sensor device fixed to the monitoring target area has a fixed sensor range (an angle of view or the like), for example, because it is possible to acquire a road shape or a position of a road marking included in the acquired image, it is possible to ascertain a road situation or a movement situation of the mobile object moving on the road with higher accuracy. According to the embodiment, maintenance can be performed more appropriately and quickly for the area monitoring system 1. Thereby, infrastructure facilities can be operated appropriately. According to the embodiment, traffic management using a radio wave sensor and an optical sensor can be performed. Therefore, the optimal types and number of sensors can be installed for each monitoring target area in

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consideration of the characteristics of each sensor. Therefore, it is possible to more appropriately ascertain the situation of the entire region with infrastructure facilities.

While preferred embodiments of the invention have been described and illustrated above, it should be understood that these are exemplary of the invention and are not to be considered as limiting. Additions, omissions, substitutions, and other modifications can be made without departing from the spirit or scope of the present invention. Accordingly, the invention is not to be considered as being limited by the foregoing description, and is only limited by the scope of the appended claims.

What is claimed is:

1. An area monitoring system comprising:
 - a plurality of sensor devices each installed in a monitoring target area and configured to detect a mobile object moving in the monitoring target area; and
 - a determiner configured to determine whether or not mobile objects detected by the plurality of sensor devices are the same physical object,
 wherein the plurality of sensor devices include at least a radio wave sensor and an optical sensor, and
 wherein the determiner determines whether or not mobile objects detected by the radio wave sensor and the optical sensor are the same physical object on the basis of information of at least one of feature quantities regarding movements of the mobile objects and reflection intensities corresponding to attributes of the mobile objects when the mobile objects have been detected by the radio wave sensor and the optical sensor.
2. The area monitoring system according to claim 1, wherein the determiner determines whether or not the mobile objects are the same physical object on the basis of information of at least one of deviations in positions of the mobile objects, speed distributions, and reflection intensities corresponding to the attributes of the mobile objects when the mobile objects move outside of a sensor range after being detected by the optical sensor and the radio wave sensor is located at a movement destination.
3. The area monitoring system according to claim 1, wherein the determiner determines whether or not the mobile objects are the same physical object on the basis of information of at least one of deviations in positions of the mobile objects, speed distributions, and reflection intensities corresponding to the attributes of the mobile objects when the mobile objects move outside of a sensor range after being detected by the radio wave sensor and the optical sensor is located at a movement destination.
4. The area monitoring system according to claim 1, further comprising a manager configured to manage a situation of the monitoring target area,

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wherein, when the mobile object is a vehicle and there is a physical object approaching the vehicle, the manager issues an action instruction for the vehicle.

5. The area monitoring system according to claim 4, wherein the manager differentiates content of the action instruction for the vehicle in accordance with a degree of influence on contact of a physical object approaching the vehicle.
6. The area monitoring system according to claim 4, wherein, when an information providing device is set in the monitoring target area, the manager causes an image or sound indicating content of the action instruction to be output to the information providing device.
7. The area monitoring system according to claim 6, wherein, when an image sensor is included in the plurality of sensor devices, the manager causes information about an attribute of a physical object approaching the vehicle included in an image captured by the image sensor to be output to the information providing device.
8. The area monitoring system according to claim 4, wherein the manager outputs information for causing the vehicle to execute one or both of speed control and steering control corresponding to content of the action instruction for the vehicle to the vehicle.
9. The area monitoring system according to claim 1, wherein the radio wave sensor is installed in a segment where curvature of a road included in the monitoring target area is less than a threshold value and the optical sensor is installed in a segment where the curvature of the road is greater than or equal to the threshold value.
10. An area monitoring method comprising:
 - determining, by a computer, whether or not mobile objects detected by a plurality of sensor devices each installed in a monitoring target area and configured to detect a mobile object moving in the monitoring target area are the same physical object on the basis of detection results of the plurality of sensor devices, wherein the plurality of sensor devices include at least a radio wave sensor and an optical sensor, and
 - determining, by the computer, whether or not mobile objects detected by the radio wave sensor and the optical sensor are the same physical object on the basis of information of at least one of feature quantities regarding movements of positions of the mobile objects and reflection intensities corresponding to attributes of the mobile objects when the mobile objects have been detected by the radio wave sensor and the optical sensor.

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